Investigating Suspected Illegal File Transfer from Memory Evidence

Case Study 3 Individual memory forensics investigation (Volatility 2)

1. Case reference and student identification

Student full name	Abubakar Mustapha Sadiq
Registration number	2025/FWSD/11202
Course code	SBT-DF204 Computer Forensics Case Studies
Case title	Investigating Suspected Illegal File Transfer From Memory Evidence
Evidence examined	Windows memory image memdumpWin7.mem (1,073,676,288 bytes)
Examination date	18 October 2026 (tool runs between 11:34 and 12:28 UTC)
Examiner workstation	Kali Linux; Volatility Foundation Volatility Framework 2.6.1 run with Python 2
Submission file name	SBTDF204_CaseStudy3_2025FWSD11202.docx (PDF copy of the same report)
Supporting archive (optional)	SBTDF204_CaseStudy3_Evidence_2025FWSD11202.zip: tool output files, screenshots and case log. It does **not** contain the memory image.

Declaration. This report is my own work. The analysis, wording and screenshots are mine, produced on my own Kali Linux workstation. The memory image was used only as authorised course material and was examined offline.

2. Executive conclusion

What the memory image supports. The image is of a Windows 7 SP1 32-bit system, captured at **2019-01-06 15:09:06 UTC**. It shows that, in an interactive session of the local account **IEUser**, cmd.exe (PID 3996, started 15:06:11 UTC by explorer.exe) held a command history of three commands in this order: ipconfig, cd Downloads and copy secret_file.docx F:. Its console screen buffer shows the prompt C:\Users\IEUser\Downloads> followed by the copy command and the text 1 file(s) copied. The registry assigns drive letter **F:** to a **USB mass-storage device** (USBSTOR, vendor General, product UDisk, revision 5.00) that Windows had registered by 15:02:54 UTC, and IEUser's Explorer recorded F:\ as its most recently used volume at 15:07:03 UTC. Memory strings tie the file name secret_file.docx to http://192.168.56.5/secret_file.docx, to Chrome download temporary names (.crdownload) and to the path C:\Users\IEUser\Downloads\secret_file.docx. The only two network records involving a remote host are closed TCP connections from this machine (192.168.56.8) to **192.168.56.5 on port 80**.

Supported conclusion. The evidence is **consistent with** a file named secret_file.docx, obtained over HTTP from 192.168.56.5 into IEUser's Downloads folder, being copied to a USB drive by a command typed at the console of IEUser's interactive session shortly before the memory was captured.

What it does not prove. It does not show the content, ownership or sensitivity of the file, or that the copy was unauthorised or illegal; the name is only a name. The success message is screen text, not an independent record of a file

on F:. No time is recorded for any single command. The closed connections have no identified owner process. The activity is attributable to an **account session (IEUser)**, not to a named person (Section 7).

Proposition	Confidence	Main basis
The command copy secret_file.docx F: was entered in an interactive cmd.exe running as IEUser	High	cmdscan history, consoles screen buffer, process token (IEUser SID), parent explorer.exe in session 1, UserAssist time
The copy completed as displayed (1 file(s) copied.)	Moderate	Console screen text only; no file object or content from F: was recovered
F: was a USB mass-storage device	Moderate to high	MountedDevices value for \DosDevices\F:, USBSTOR and DeviceClasses entries, MountPoints2, ShellBags volume F:\
The file came from 192.168.56.5 via Chrome	Moderate	Strings (URL, omnibox searches, .crdownload) plus closed port-80 connections to the same host; strings are undated and the sockets have no owner
Order of events within 15:02–15:09 UTC	Moderate	Registry and process times agree on one clock, but the early-boot clock was inconsistent and the commands carry no times (Section 5.2)
The activity identifies a particular person, intent or illegality	Low / not supported	Shared lab-style account, an SSH service on the host, a capture tool run in the same session

Overall confidence: moderate. At least three limitations affect this level: (1) the console buffer shows what was displayed, not what was written to F:; (2) no per-command timestamps and an inconsistent early-boot clock; (3) closed network connections with owner PID −1 cannot be linked to a process. The full list is in Section 5.8.

3. Evidence acquisition and integrity

The evidence is a Windows physical-memory image. The scenario text of the assessment brief calls it a "Windows 10 memory image", but the file name (memdumpWin7.mem) and the course materials say Windows 7. The operating system was therefore taken from the image itself (Section 4); the profile selected from imageinfo is Windows 7 SP1 x86.

Original file name	memdumpWin7.mem
Source	Course material for SBT-DF204 Case Study 3, downloaded to ~/Downloads on the examiner's Kali workstation
File size	1,073,676,288 bytes (1 GiB less 64 KiB)
SHA-256 (original)	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
SHA-256 (working copy)	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e (identical)
Read-only original	~/SBTDF204_CaseStudy3/01_original_evidence/memdumpWin7.mem (mode 444, -r--r--r--)
Working copy	~/SBTDF204_CaseStudy3/02_working_copy/memdumpWin7.mem (all Volatility runs use this file)
Tool output	~/SBTDF204_CaseStudy3/03_volatility_output/ (each file starts with the plugin, profile, exact command and UTC run time)
Tools	cp, chmod, stat, sha256sum, find, strings; Volatility 2.6.1 (official GitHub repository); Python 2.7 with pycryptodome 3.24.1

Method. The case folder was created first, with separate folders for original evidence, working copy, Volatility output, screenshots, notes and submission (Figure S01). The image was copied with cp -p into 01_original_evidence/ and made read-only with chmod 444. The first copy attempt used a placeholder instead of the real path and failed, so no file was created (total 0 in Figure S01); the corrected command succeeded (Figure S02). The size was read with stat and the SHA-256 with sha256sum. A working copy was made and hashed; the two hashes are identical (Figure S02). Every analysis command was run on the working copy only. Each action was logged with a UTC time in 05_notes/case_log.txt.

Constraints from the brief, and how they were met.

- The memory image was **not** uploaded anywhere. Only tool output, screenshots and the case log are packaged for submission.
- No remote IP address, host, account, service or person named in the output (for example 192.168.56.5) was contacted. The examination was offline; the host and port numbers appear only as recorded evidence.
- No recovered file was executed, and no password or hash cracking was attempted.
- The screenshots show the terminal only. They contain no unrelated personal data, so no redaction was needed.

Chain of custody. One examiner handled the file. It was downloaded once, hashed on arrival, copied once into the read-only evidence folder and once into the working folder; the hashes matched after each copy.

4. Method and profile selection

4.1 Tooling

Volatility 2.6.1 was cloned from the official repository (volatilityfoundation/volatility) into ~/tools/volatility and run with python2 vol.py (Figure S03). Registry plugins such as shellbags, userassist and getsids did not load at first because the Python 2 module Crypto.Hash was missing; installing pycryptodome 3.24.1 for Python 2 fixed this (Figure S07). The distorm3 module is also absent, which disables only the malware, ssdt and API-hook plugins. Those plugins were not needed. The "Failed to import" notices in the saved output files are these tool-import messages and are not findings.

4.2 Profile selection and validation

imageinfo was run without a profile. Its first run was terminated by the shell (zsh: killed, probably memory exhaustion in the VM), leaving only a header in 01_imageinfo.txt; this file was not used (Figures S04, S05). After closing other applications it was re-run and saved as 01b_imageinfo.txt (Figure S06). Suggested profiles: Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86. kdbgscan then confirmed the kernel build string 7601.18939.x86fre.win7sp1_gdr.15, with the owner-tag check true and 54 processes on PsActiveProcessHead (Figure S08).

Win7SP1x86_23418 was chosen as the working profile. It was validated by checking that several independent plugins returned coherent results with it: pslist and psscan (a complete process tree with sensible parents and sessions), netscan (socket owners resolve to process names), hivelist (hive paths resolve, including IEUser's ntuser.dat), printkey and hivedump (readable registry keys), and consoles (readable text). **Limit:** Win7SP1x86_24000 has the same kernel build and was not run for comparison, so the choice rests on the first suggestion plus this coherence check.

4.3 Reading the evidence

Findings are written as **observed fact** (what the tool output shows) and **interpretation and limit** (what it may mean and what it cannot show). Evidence IDs (E01–E16) refer to the evidence log in Appendix D; figure IDs (S01–S31) refer to the screenshots in Appendix A. Times are UTC. All times come from the system clock stored in the image, which is discussed in Section 5.2.

5. Findings

5.1 Q1Evidence preservation (E01, E16)

Observed. The original image is stored read-only with SHA-256 500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e (1,073,676,288 bytes). The working copy has the same hash. All plugins were run on the working copy (the command line in every output header names 02_working_copy/memdumpWin7.mem). Figures S01 and S02; details in Section 3.

Interpretation and limit. The matching hashes show that the working copy is a bit-identical duplicate of the file as downloaded. They do not prove that the downloaded file is identical to the lecturer's original, because no reference hash was supplied. The case log records each step with a UTC time (Figures S02, S15, S29, S31).

5.2 Q2Profile, image time and time zone (E02, E03)

Item	Value (source: imageinfo unless stated)
Chosen profile	Win7SP1x86_23418 (Windows 7 SP1, 32-bit, kernel 6.1.7601)
Other suggestions	Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
Build string (kdbgscan)	7601.18939.x86fre.win7sp1_gdr.15; KDBG owner-tag check true; PsActiveProcessHead 0x8298b4f0 (54 processes)
Address space	IA32PagedMemoryPae over FileAddressSpace; PAE; DTB 0x185000
KDBG / KPCR / KUSER_SHARED_DATA	0x82972c68 / 0x82973d00 / 0xffdf0000; 1 processor; service pack 1
Image date and time	**2019-01-06 15:09:06 UTC**
Image local date and time	2019-01-06 07:09:06 **−0800**

Figures S06 and S08. The profile was validated with multiple plugins as described in Section 4.2.

Time zone

Observed. imageinfo derives the local time from live kernel data and reports an offset of **UTC−08:00** (07:09:06 local when 15:09:06 UTC). The registry key ControlSet001\Control\TimeZoneInformation exists (hivedump lists it) but printkey returned no values for it, with or without the SYSTEM hive offset (-o 0x87c1c008), so the zone name was not recovered from the registry (Figures S09, S10). The key's last-written time is 2019-01-06 **16:31:39 UTC**.

Interpretation and limit. An offset of −08:00 corresponds to Pacific Standard Time in January, but that name is an inference; only the offset is observed. The key's last-written time (16:31:39) is **later than the image time** (15:09:06), so registry last-write times cannot all share one clock with the image time. This is part of the clock problem below.

Clock inconsistency in early-boot process times

Observed. In pslist, the first processes (System, smss.exe, both csrss.exe, wininit.exe, winlogon.exe, services.exe, lsass.exe, lsm.exe, svchost.exe 552 and VBoxService.exe 616) are stamped **18:02:44–18:02:45 UTC**. The next processes (svchost.exe 672, 724, 828) are stamped **15:02:47**, and everything afterwards runs from 15:02:47 to 15:09:07 UTC, which fits the image time of 15:09:06 (Figures S08, S12, S17). The volume GUID {421e7e52-11dd-11e9-b3cf-08002709e15d} contains a creation time that decodes (version-1 UUID timestamp, checked with Python) to **2019-01-06 18:02:40.6 UTC** (Figure S23), which is on the early clock.

Interpretation and limit. The step is about three hours back (18:02:45 → 15:02:47), between VBoxService.exe and the next svchost.exe. A plausible explanation is that the guest clock was synchronised with the host shortly after start-up (VBoxService.exe, VBoxTray.exe and the Hyper-V integration processes vmicsvc.exe are all present). This is a hypothesis, not a finding. The timeline therefore uses only times stamped after the step (15:02:47 onward), which agree with the image time. Events stamped on the early clock are used only as context. Registry last-write times from earlier sessions (for example 14:32:25 for the CD-ROM MountPoints2 entries and 14:49:40 for the Downloads shell item) are treated as indicators and are not sequenced against process times.

dllhost.exe (PID 2228) is stamped 15:09:07, one second **after** the recorded image time, which shows that capture is not instantaneous and that the image time and the last process times can differ by about a second.

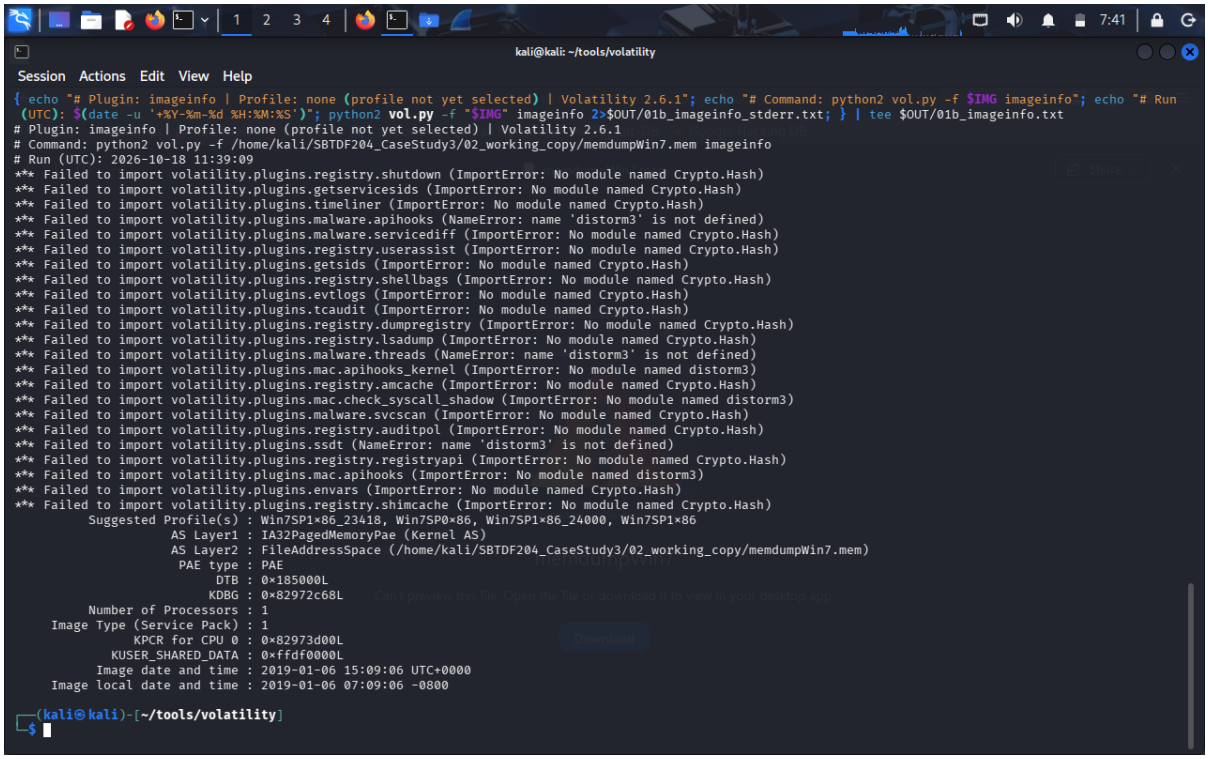


Figure S06*Successful* imageinfo *run: suggested profiles, address spaces, KDBG, image date/time and local offset.*

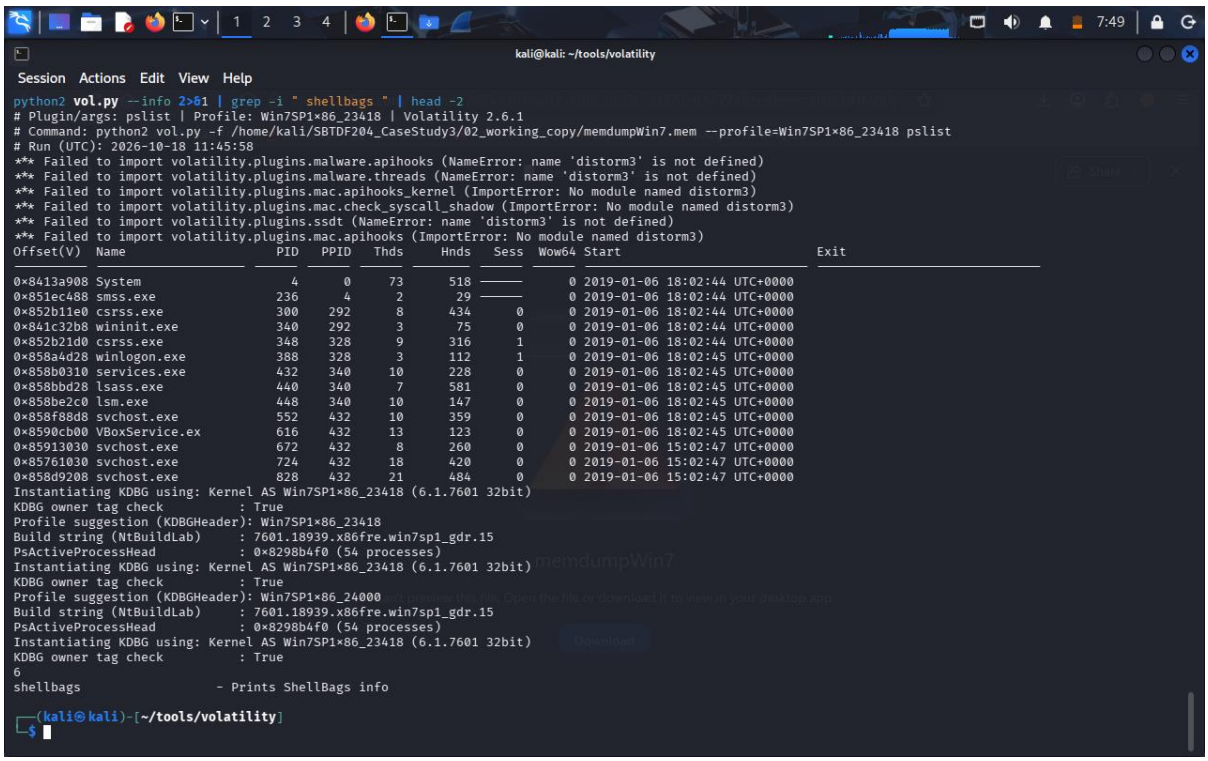


Figure S08pslist *start of output (early-boot times 18:02:44–45, later 15:02:47) with* kdbgscan *build confirmation.*

5.3 Q3User and account artefacts (E04, E08)

Observed. getsids was run for PIDs 2524, 3996, 2388 and 2596. The cmd.exe (3996) token starts with the user SID S-1-5-21-1716914095-909560446-1177810406-1000, labelled **IEUser**, followed by group S-1-5-21-…-513 (labelled "Domain Users" by the plugin), Everyone, Local Account / Administrators (S-1-5-114, S-1-5-32-544), Users, **Interactive (S-1-5-4)**, **Console Logon (S-1-2-1)**, Authenticated Users, This Organization, Local Account (S-1-5-113), a logon-session SID (S-1-5-5-0-61708), Local (S-1-2-0), NTLM Authentication (S-1-5-64-10) and **Medium Mandatory Level (S-1-16-8192)**. There is no Network (S-1-5-2) group. The FTK Imager.exe (2596) token has the same user SID and **High Mandatory Level (S-1-16-12288)** (Figures S10, S11).

The SAM Names key lists the local accounts **Administrator, Guest, IEUser, sshd and sshd_server**; the key was last updated on 2015-09-21 09:50:50 UTC, so these accounts date from the image build (Figure S11). hivelist shows loaded hives for IEUser (ntuser.dat, virtual 0x8b5d39b0, physical 0x120049b0; UsrClass.dat, 0xa620b008) and for sshd_server (ntuser.dat 0xa65269c8; UsrClass.dat 0xa6539260), plus SYSTEM 0x87c1c008, SOFTWARE 0x9711d9c8, SAM 0x98209518, SECURITY 0x97104008 and DEFAULT 0x981f9008 (Figures S13–S15).

Hive (path)	Virtual / physical offset	Use in this report
\REGISTRY\MACHINE\SYSTEM	0x87c1c008 / 0x27fa3008	DeviceClasses, USBSTOR, MountedDevices, TimeZoneInformation
\SystemRoot\System32\Config\SOFTWARE	0x9711d9c8 / 0x1e6a09c8	ProfileList attempt (no data recovered)
\SystemRoot\System32\Config\SAM	0x98209518 / 0x1f3a7518	Local account names
\??\C:\Users\IEUser\ntuser.dat	0x8b5d39b0 / 0x120049b0	MountPoints2, UserAssist
…\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	0xa620b008 / 0x1997b008	ShellBags

Hive (path)	Virtual / physical offset	Use in this report
\??\C:\Users\sshd_server\ntuser.dat and UsrClass.dat	0xa65269c8 / 0x17ad09c8; 0xa6539260 / 0x0f690260	Shows the sshd_server service-account profile was loaded

Interpretation and limit. The cmd.exe and FTK Imager.exe processes ran under the **IEUser profile**. An account or SID identifies a profile, not a person: IEUser is a local account that, with its 2015 build date, looks like a shared or default lab account, and nothing in the memory shows who authenticated or who was at the keyboard. The ProfileList key, which would map the SID to its profile path and load time, could not be recovered (empty output in Figures S10 and S25), so the SID-to-path link rests on the loaded C:\Users\IEUser\ntuser.dat hive and the C:\Users\IEUser> console prompt.

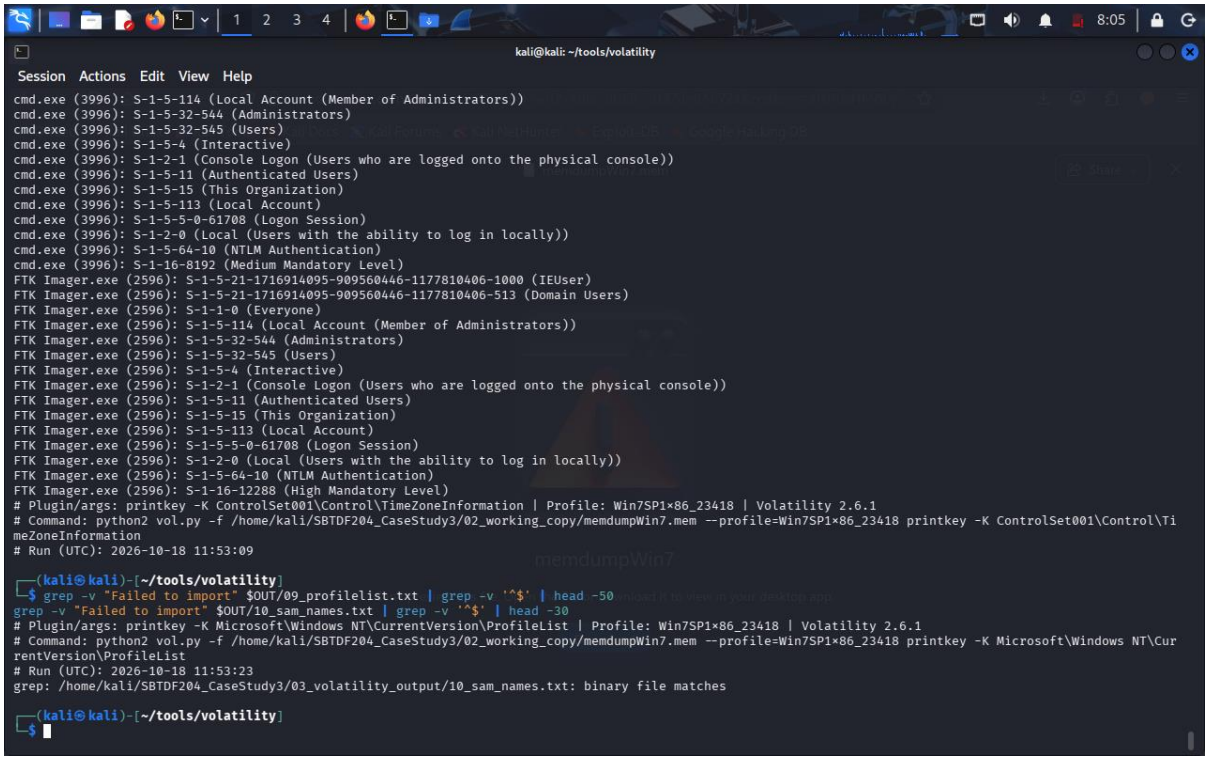


Figure S10getsids *tokens for* cmd.exe *and* FTK Imager.exe*, with the empty* TimeZoneInformation *and* ProfileList *runs.*

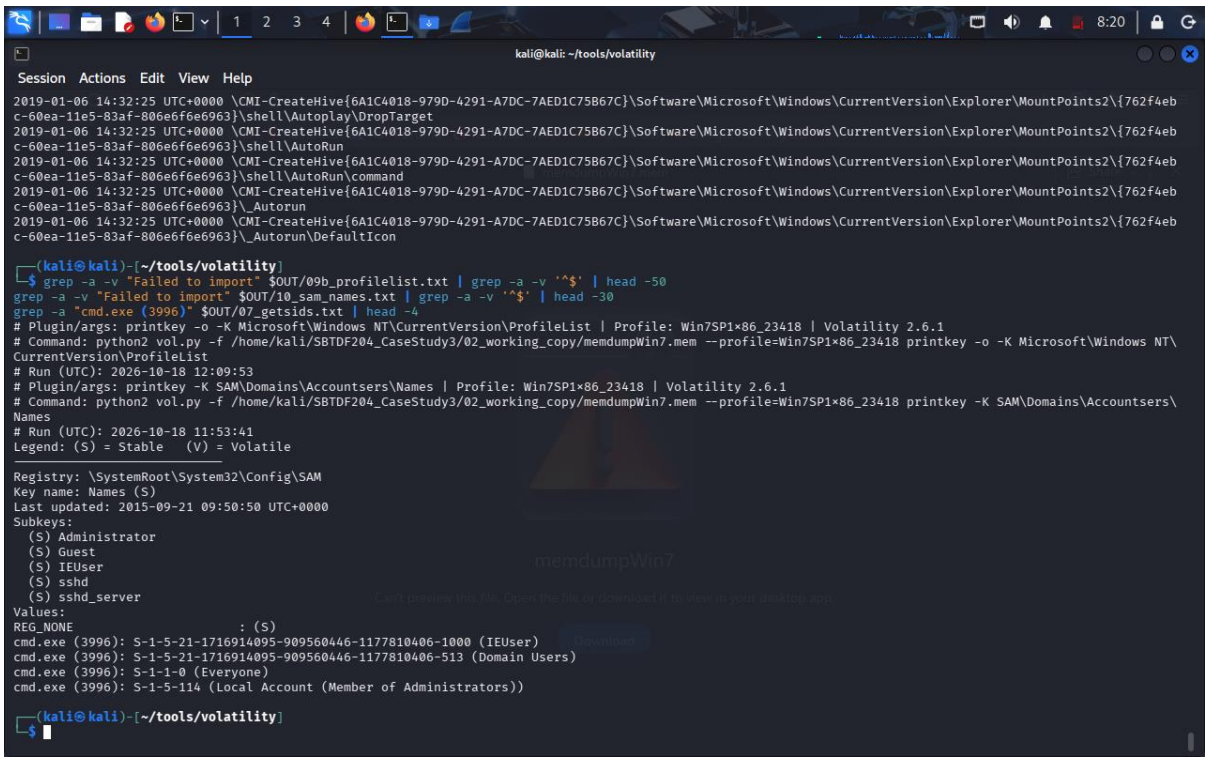


Figure S11*SAM* Names *key (five local accounts) and the first lines of the* cmd.exe *token (IEUser SID).*

5.4 Q4Processes and network (E05, E06)

Observedprocesses (pslist, cross-checked with psscan; Figures S08, S12, S17):

PID	Process	PPID	Session	Created (UTC)	Note
616	VBoxService.exe	432	0	18:02:45 (early clock)	Guest-integration service; last process before the clock step
1680 / 1984 / 2016	cygrunsrv.exe / cygrunsrv.exe / sshd.exe	432 / 1680 / 1984	0	15:02:48–49	OpenSSH server service; conhost.exe 2000 is its console
2524	explorer.exe	2500	1	15:02:54	IEUser's interactive shell; parent 2500 is not in the process list
3748	msiexec.exe	432	0	15:03:34	Windows Installer service
2388	chrome.exe	2524	1	15:04:30	Child processes 1280, 2380 (15:04:30), 2912 (15:04:32), 976 (15:06:23)
3996	cmd.exe	2524	1	**15:06:11**	Console of the commands; owned by conhost.exe 3988
3988	conhost.exe	348	1	15:06:11	Console host for cmd.exe 3996 (parent is the session-1 csrss.exe)
2596	FTK Imager.exe	2524	1	15:07:15	Memory-capture tool, High integrity
2228	dllhost.exe	552	1	15:09:07	Created one second after the image time

Observednetwork (netscan; Figures S13–S15, S17):

Offset	Proto	Local → remote	State / owner	PID	Created
0x3e043b10	TCPv4	192.168.56.8:49177 → 192.168.56.5:80	**CLOSED**	−1	—
0x3fcba008	TCPv4	192.168.56.8:49178 → 192.168.56.5:80	**CLOSED**	−1	—
0x3ee1aa88	TCPv4	0.0.0.0:22 (and :::22)	LISTENING sshd.exe	2016	—
—	TCPv4	0.0.0.0:445; 192.168.56.8:139	LISTENING System	4	—
0x3e081f50	UDPv4	192.168.56.8:56647	chrome.exe	2388	15:08:32
0x3e2863d8	UDPv4	0.0.0.0:0	VBoxService.exe	616	15:08:35

Interpretation and limit. netscan shows **no ESTABLISHED connection**. The only connections to a remote host are the two closed TCP endpoints to 192.168.56.5 on port 80 (HTTP), from the local address 192.168.56.8, which ipconfig in the console also shows (Local Area Connection 3, mask 255.255.255.0, no default gateway). Both addresses are in a private 192.168.56.0/24 network, typical of a host-only lab network. The owner PID is −1, so these records **cannot be tied to chrome.exe or to any process**, and they carry no time. They are consistent with, but do not prove, that a program on this machine made HTTP requests to 192.168.56.5. The host was not contacted during this examination. The SSH service listens on port 22 (Section 7).

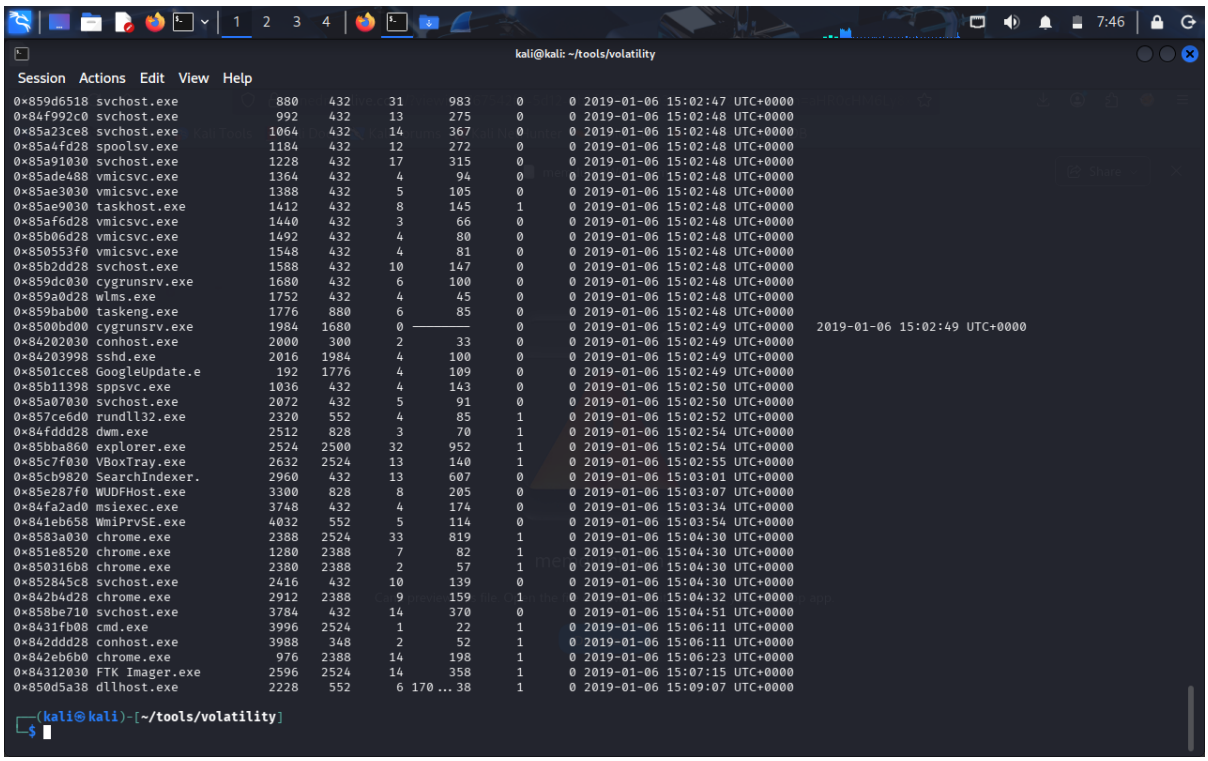


Figure S12pslist *(later part): process creation times from 15:02:47 to 15:09:07 UTC, with* explorer.exe*,* chrome.exe*,* cmd.exe*,* FTK Imager.exe*.*

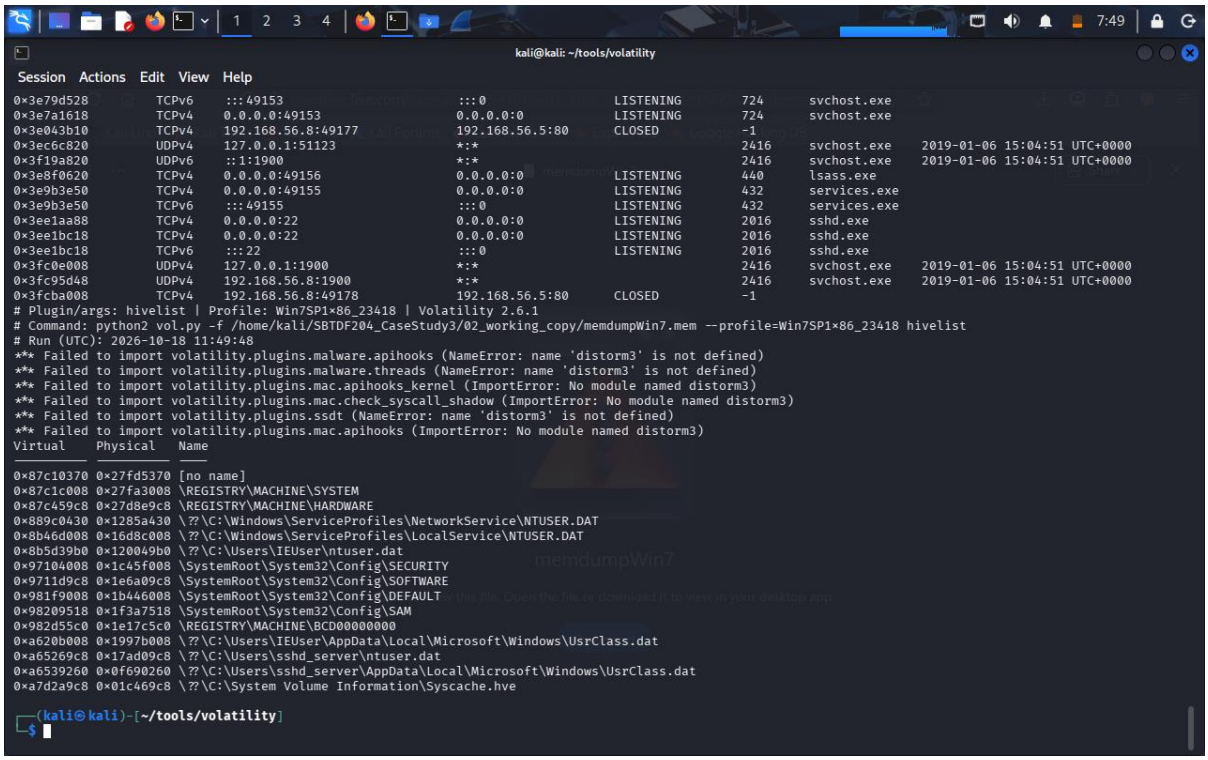


Figure S14netscan *end: two CLOSED TCP rows to 192.168.56.5:80 (owner −1) and* sshd.exe *listening on port 22;* hivelist*.*

5.5 Q5Command history and console evidence (E07, E08, E09, E14)

Observed. cmdscan found a command history in the console hosted by conhost.exe **3988**, belonging to cmd.exe (Figures S16–S19):

```
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
```

```
CommandCount: 3 LastAdded: 2 LastDisplayed: 2   FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
```

consoles links this history to AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c, OriginalTitle: %SystemRoot%\system32\cmd.exe, and shows the screen buffer (Screen 0xf63b8 X:80 Y:300):

```
C:\Users\IEUser>ipconfig        ... Local Area Connection 3: IPv4 Address . . : 192.168.56.8
                                Subnet Mask . . . . . : 255.255.255.0   Default Gateway : (blank)
C:\Users\IEUser>cd Downloads
C:\Users\IEUser\Downloads>copy secret_file.docx F:
       1 file(s) copied.
C:\Users\IEUser\Downloads>
```

The cmd.exe process (PID 3996) was created at **15:06:11 UTC** with parent explorer.exe 2524 in session 1; UserAssist records cmd.exe last launched at 15:06:11 (count 10; the counters are lifetime values). Its token has the Interactive and Console Logon SIDs and the IEUser SID (Section 5.3). A second console, conhost.exe 2000, belongs to the OpenSSH service (sshd.exe 2016) and has CommandCount: 0. Entries #22, #25, #36 and #37 (?, ?????) lie beyond CommandCount: 3 and are residual memory, not commands; they are not used.

Supporting traces for the file (strings on the working copy and filescan; Figures S30, S31). The file name occurs in these forms, among others: http://192.168.56.5/secret_file.docx (about 38 times as ASCII, and 192.168.56.5/secret_file.docx 58 times as UTF-16); Chrome omnibox search URLs …search?q=192.168.56.5%2Fsecret_file.docx&sourceid=chrome…; secret_file.docx.crdownload and secret_file.docx.crdownload….tmp; and C:\Users\IEUser\Downloads\secret_file.docx (16 times) with file:C:/Users/IEUser/Downloads/secret_file.docx (19 times). filescan shows the object \Device\HarddiskVolume1\Users\IEUser\Downloads and Links\Downloads.lnk. Variant strings with other digits (196.128.56.5…, 192.128.56.5…) also occur; they could be mistyped input or damaged copies, no connection to those addresses appears in netscan, and they are not relied on.

Interpretation. The history shows, in order, that the user checked the network configuration, moved to the Downloads folder and copied secret_file.docx to drive F:. The console displayed 1 file(s) copied. The strings are consistent with the file having been requested over HTTP from 192.168.56.5 by Chrome and saved in IEUser's Downloads folder before the copy. The copy could only have shown "1 file(s) copied." if a file of that name existed in the current directory at the moment it ran.

Limits on proving the transfer.

- The command history and the console screen buffer are two views of the same console; they are not independent confirmations.
- "1 file(s) copied." is text that was displayed. No file object for secret_file.docx on F: and no content from the device were recovered, and F: itself is identified only through the registry (Section 5.6).
- The history records order, not time. The copy ran between the start of cmd.exe (15:06:11) and the capture (15:09:06). The UserAssist time is when the console was launched, not when a command ran.

- The strings carry no timestamps, their counts are remnants in memory pages and not counts of events, and they cannot show when or by which process the file was downloaded.

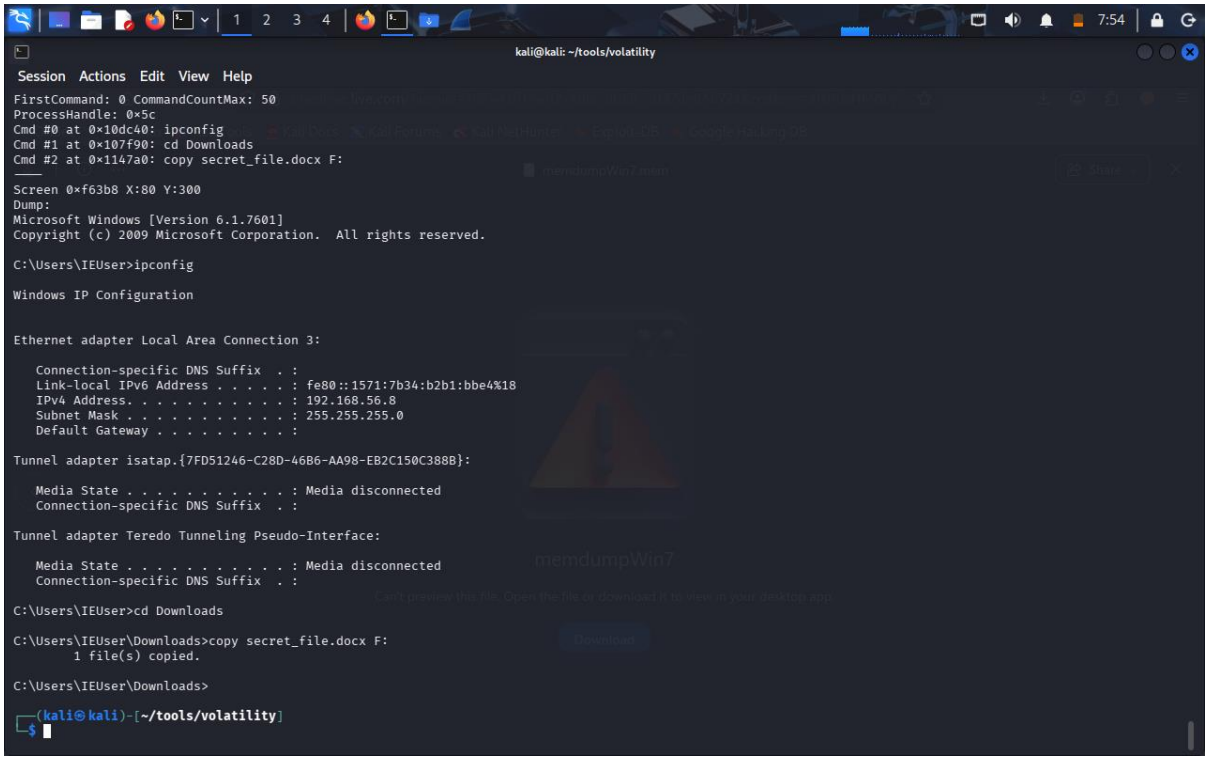


Figure S16consoles *screen buffer for console 3988:* ipconfig, cd Downloads, copy secret_file.docx F: *and* 1 file(s) copied.

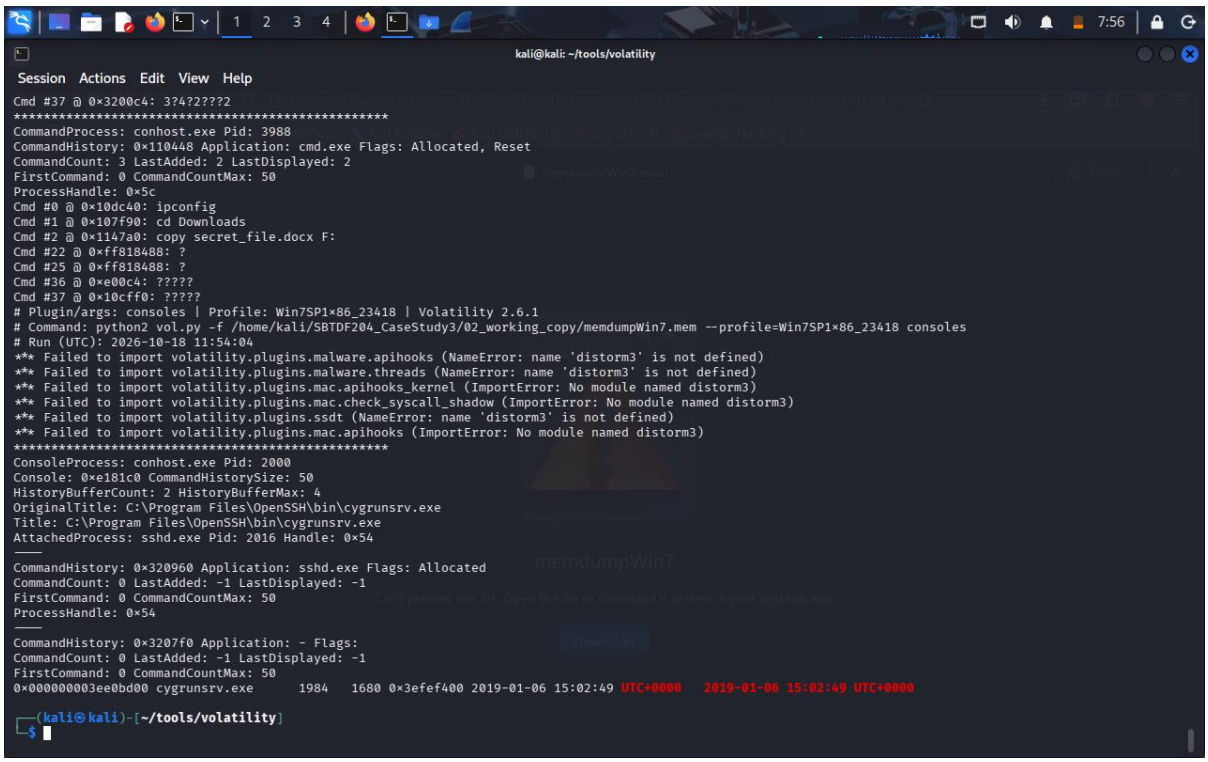


Figure S18cmdscan *(history 0x110448, three commands,* LastAdded: 2*) and the start of* consoles *with the OpenSSH service console.*

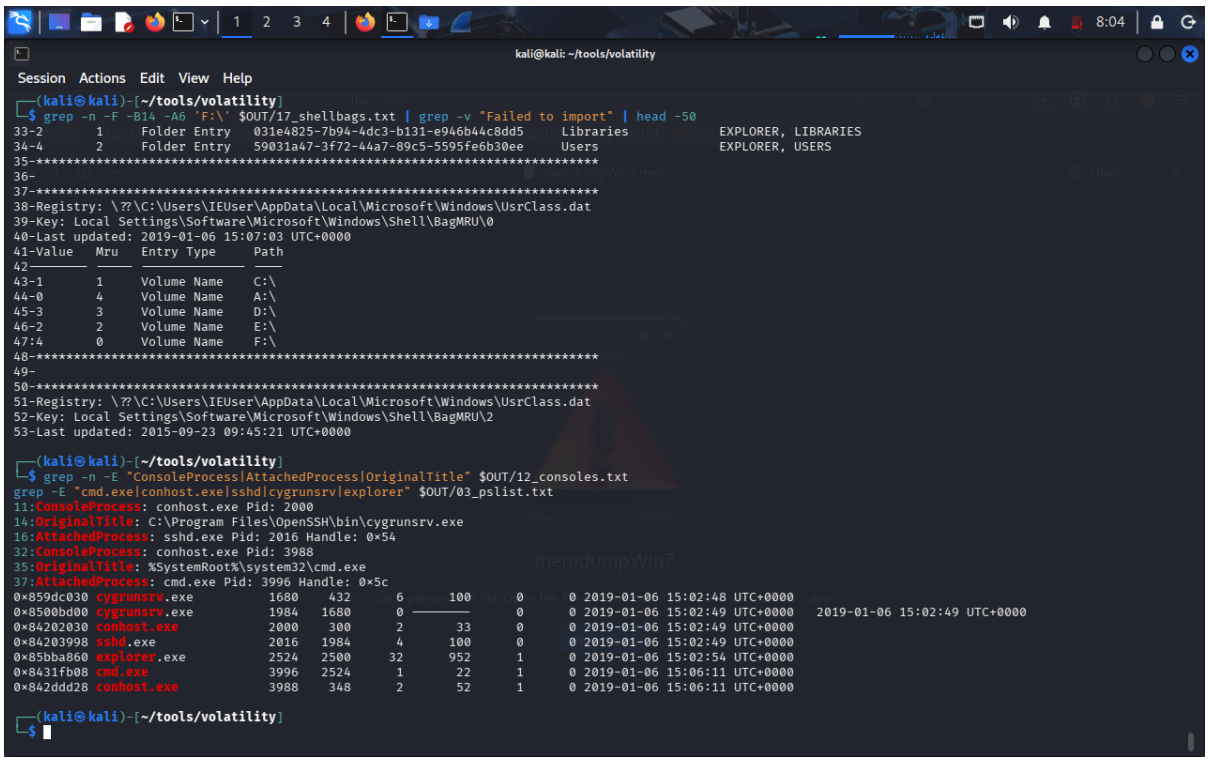


Figure S19*Console owner lines,* pslist *rows for* cmd.exe/conhost.exe *(created 15:06:11, parent 2524) and the BagMRU volume list.*

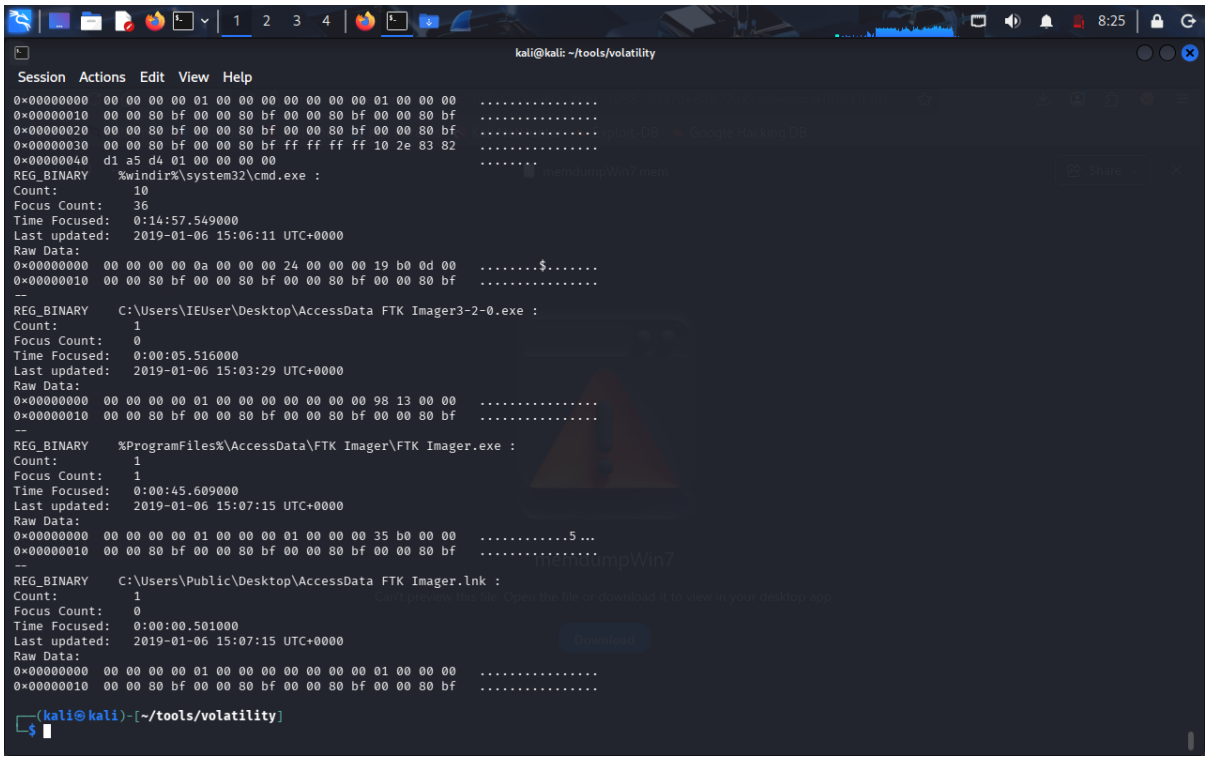


Figure S20userassist*:* cmd.exe *last updated 15:06:11; FTK Imager installer 15:03:29;* FTK Imager.exe *15:07:15.*

5.6 Q6Removable media (E10, E11, E12, E15)

Artefact	Observed (UTC)
DeviceClasses disk class {53f56307-b6bf-11d0-94f2-00a0c91efb8b}	Key last updated 15:02:54. Five disk ##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f56307… VMware, Dell VIRTUAL_DISK)
Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0 and Enum\STORAGE\Volume_??_USBSTOR#…	Keys and subkeys (Device Parameters, Properties, LogConf) all last written **15:02:54**
MountedDevices (SYSTEM)	Key last updated 15:02:54. \DosDevices\F: holds _??_USBSTOR#Disk&Ven_General&Prod entries: C:, D:, E: and volume GUIDs (floppy, CD-ROM, disk)

Artefact	Observed (UTC)
MountPoints2 (IEUser NTUSER.DAT)	Key last updated 15:03:07. Subkeys F (created 15:03:06), {421e7e52-11dd-11e9 CPC\Volume\{421e7e52…} (15:03:07), and four older volume GUIDs (2015; CD-ROM er
Windows Portable Devices interfaces (SYSTEM DeviceClasses)	##?#WpdBusEnumRoot#UMB#…STORAGE#VOLUME#_??_USBSTOR#DISK&VEN_GENERA created **15:03:08**
Volume GUID timestamps (decoded with Python)	{421e7e52-…} → 2019-01-06 **18:02:40.6** (early clock). The four others → 2015-09-21 /
ShellBags volume list (IEUser UsrClass.dat)	Volume F:\ present (Section 5.7)

Interpretation. The registry assigns **F: to a USB mass-storage device (General UDisk, Rev 5.00)**. That device and its volume were registered at 15:02:54–15:03:08, about three minutes before the console started (15:06:11), so the drive was present and mounted before copy secret_file.docx F: was typed. **Limits:** key last-write times show when Windows updated the key, not when a person inserted the stick; the instance ID is not a hardware serial number; USBSTOR output file 13_usbstor.txt contained non-text bytes and was reviewed through hivedump instead; no file listing of the device is available from memory.

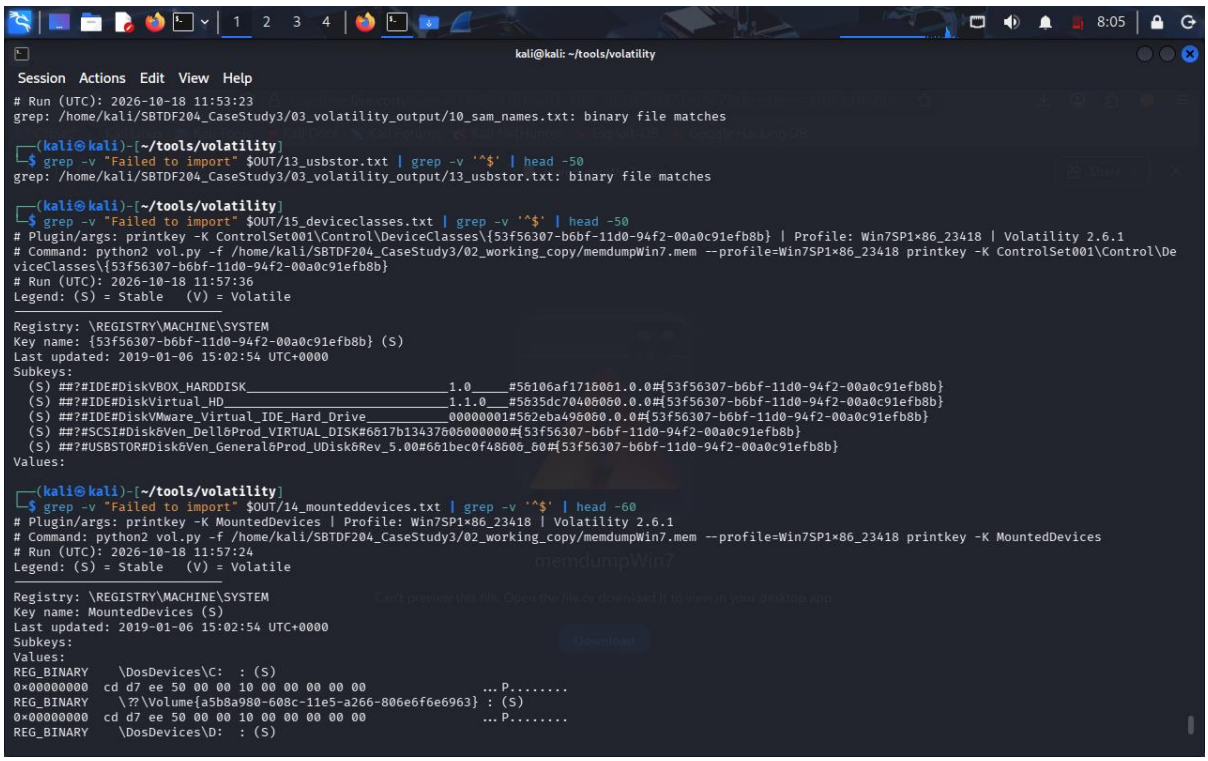


Figure S21DeviceClasses *disk class (five devices including the USB disk) and the start of* MountedDevices.

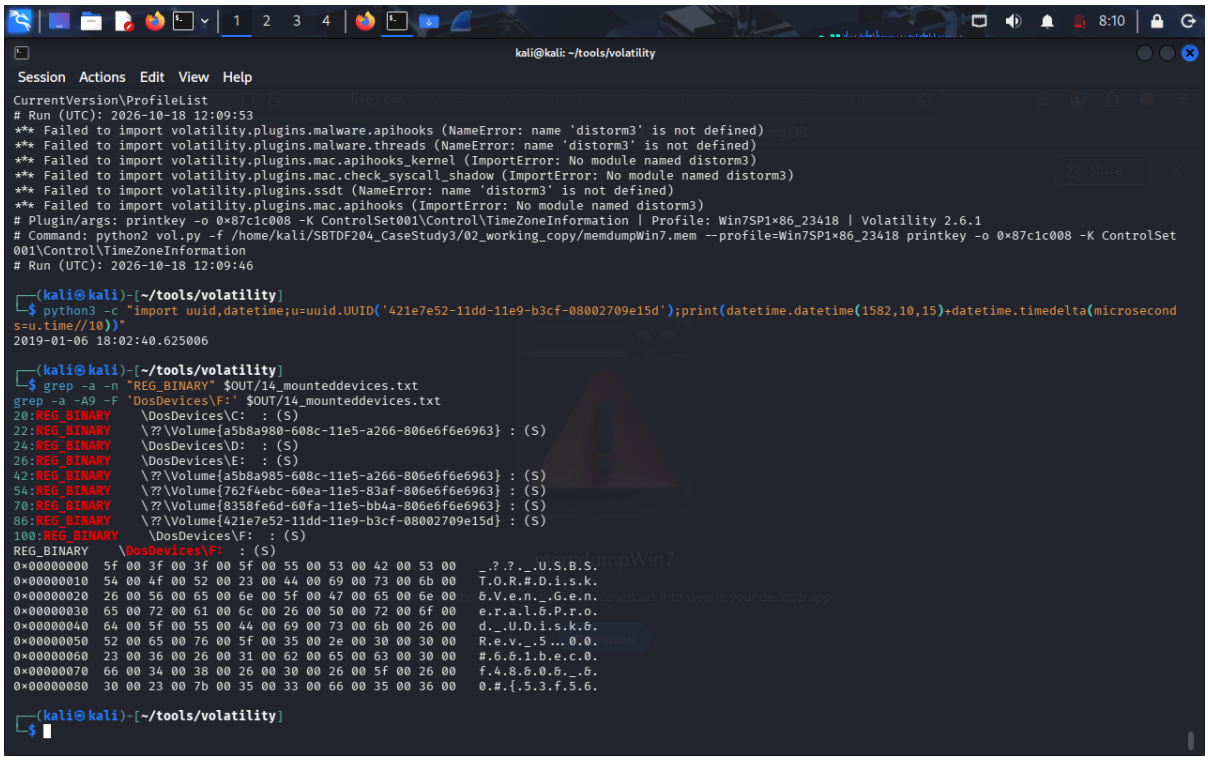


Figure S23*Volume GUID creation-time decoding and* MountedDevices*:* \DosDevices\F: *reads* _??_USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0.

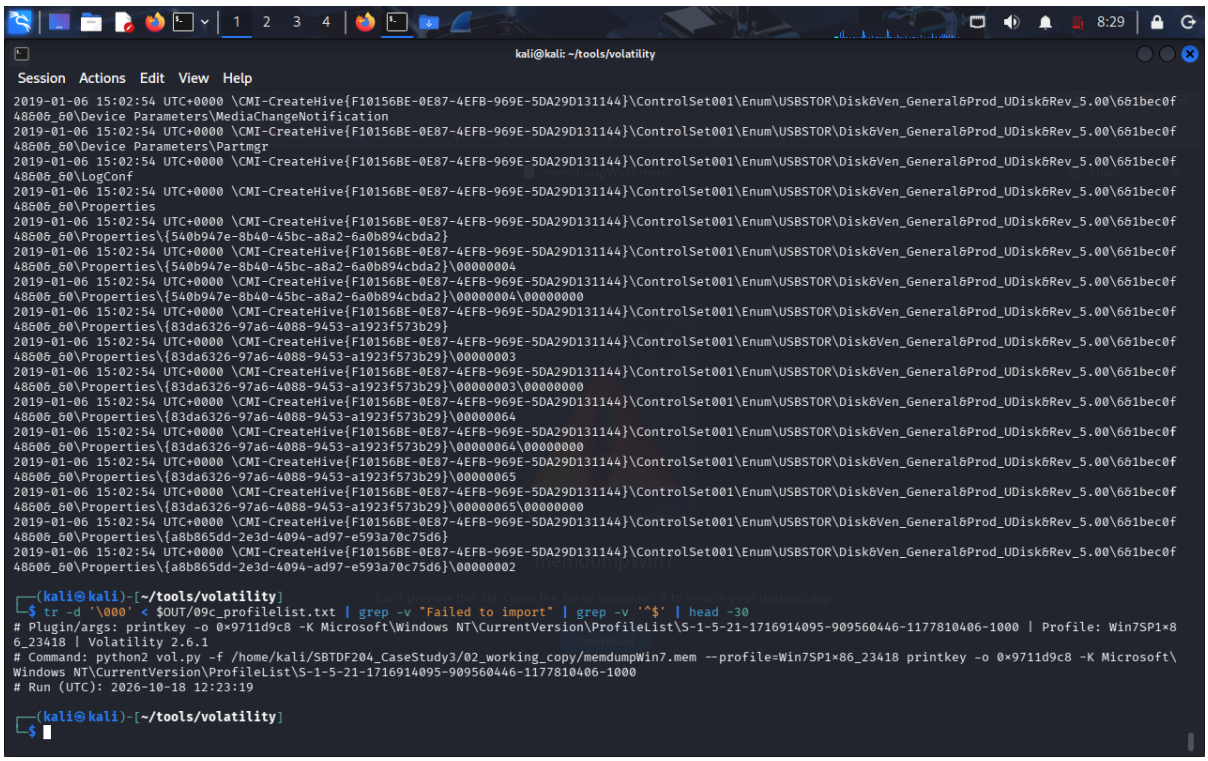


Figure S25hivedump *(SYSTEM):* Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0 *and subkeys written 15:02:54.*

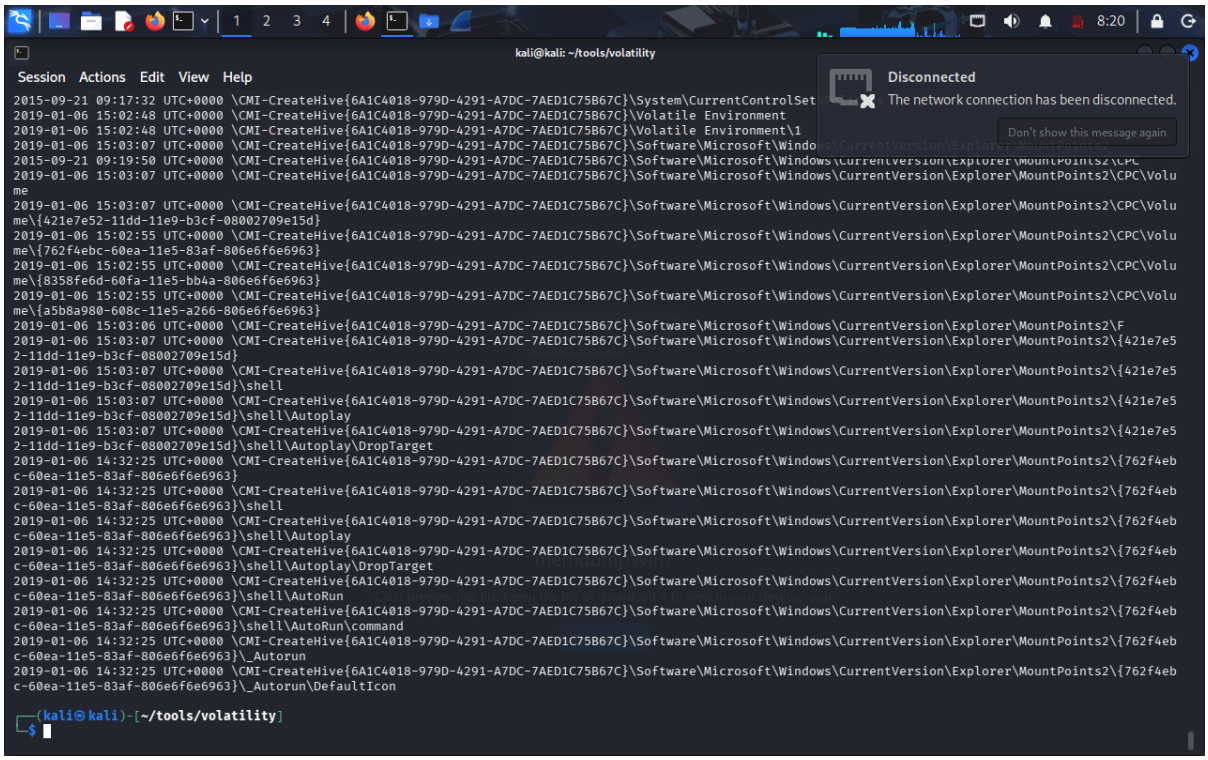


Figure S27hivedump *(IEUser* NTUSER.DAT*) for* MountPoints2*:* F *created 15:03:06;* {421e7e52-…} *created 15:03:07.*

5.7 Q7ShellBags (E13)

Observed. shellbags was run on the working copy (text and the HTML output the brief asks for, 18_shellbags.html; Figures S28, S29). In IEUser's UsrClass.dat, key Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0 (**last updated 2019-01-06 15:07:03 UTC**) lists these volumes:

Value	MRU	Entry type	Path
1	1	Volume Name	C:\
0	4	Volume Name	A:\
3	3	Volume Name	D:\
2	2	Volume Name	E:\
4	**0**	Volume Name	F:\ (MRU 0 = most recently used)

The same output has a **Downloads** folder entry (known-folder GUID 374de290-123f-4565-9164-39c4925e467b) and a short-name item DOWNLO~1 with path C:\Users\IEUser\Downloads, folder **Modified 2019-01-06 14:49:40, Created 2015-09-21 09:17:34, Accessed 2019-01-06 14:49:40 UTC**, attributes RO, DIR. Other BagMRU entries (Temp, Windows, Start Menu) have 2015 build-era dates.

Interpretation and limit. ShellBags are written by Explorer when a folder is browsed, so F:\ being the most recently used volume at 15:07:03 shows that **IEUser's Explorer navigated to F:** about 52 seconds after cmd.exe started and 12 seconds before FTK Imager.exe started. That is consistent with the user looking at the drive after the copy, but it cannot show when the copy ran or which files were seen. Two cautions apply, as the brief asks. First, the BagMRU key time is when the registry key was written, not the time a particular file was opened. Second, the dates stored inside a shell item (Modified, Created, Accessed) are the **folder's own file-system times as they were when Explorer recorded the item**; they are not the modification time of secret_file.docx and not the time the user viewed the folder. The Downloads item (14:49:40) is a snapshot taken at an earlier visit, so later changes to the folder, including a later arrival of secret_file.docx, would not appear in it. It also predates this boot's process times (from 15:02:47), which is another reason not to sequence it against them (Section 5.2).

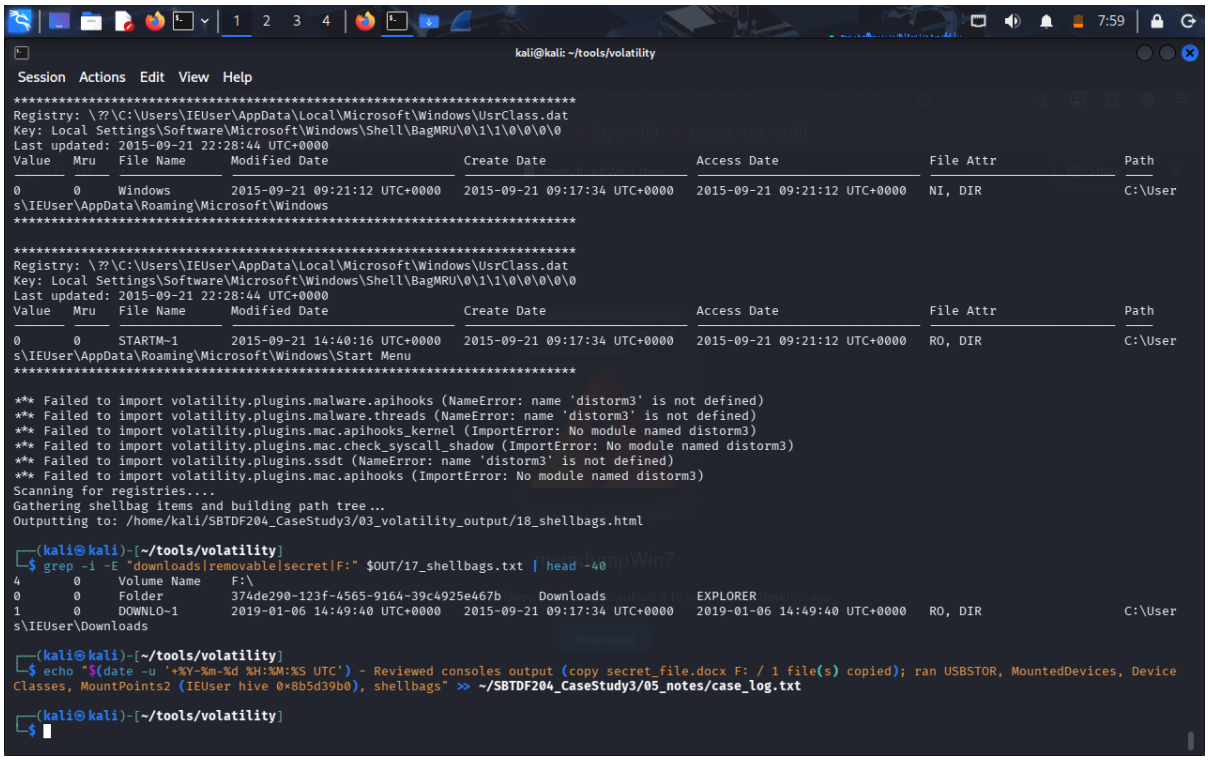


Figure S29grep *of the ShellBags text for* F:\ *and Downloads: volume* F:\ *and* DOWNLO~1 *with 14:49:40 times.*

5.8 Q8Timeline, conclusion, confidence and limitations

Timeline. The UTC timeline is in Section 6 (summary and figure) and Appendix C (full table with source artefacts, observed facts and interpretation).

Cautious conclusion. The memory evidence is consistent with a secret_file.docx that was present in IEUser's Downloads folder being copied to drive F:, a USB mass-storage device that had been mounted since about 15:03 UTC, by a command typed in an interactive cmd.exe session between 15:06:11 and 15:09:06 UTC on 2019-01-06. The console reported success. It is also consistent with the file having been fetched by Chrome from 192.168.56.5. The evidence does not establish the file's content, its sensitivity or ownership, whether the transfer was authorised, or who performed it. "Suspected illegal" remains a suspicion: nothing in memory addresses legality.

Confidence level: moderate (propositions and levels are in Section 2).

Limitations:

#	Limitation
1	**Screen text is not proof of a write.** The console shows 1 file(s) copied., but no file object or content from F: was recovered, and the history and screen buffer are the same console seen twice.
2	**No per-command times.** cmdscan and consoles record order, not time. The copy is bounded only by cmd.exe creation (15:06:11) and the image time (15:09:06). UserAssist gives the launch time of cmd.exe, not of a command.
3	**Unowned closed connections.** The two port-80 connections to 192.168.56.5 are CLOSED with owner PID −1 and no times; they cannot be linked to chrome.exe. A memory snapshot also cannot show traffic that left no trace.
4	**Clock inconsistency.** Early-boot processes are stamped three hours later (18:02:44–45) than later ones (15:02:47+); the TimeZoneInformation key was last written at 16:31:39, after the image time; shell and CD-ROM entries date from 14:32–14:49. Registry last-write times are therefore indicators, not proof of order across sessions. The cause (probably a clock synchronisation) is a hypothesis.
5	**Registry times are last-write times.** They show when a key changed, not when a device was plugged in or a folder was opened; ShellBag item dates are folder snapshots, and MountedDevices values have no time of their own.
6	**Profile is not a person.** IEUser looks like a shared lab account; an OpenSSH server (sshd, sshd_server) was running; and FTK Imager.exe ran under the same account, so the examiner's own activity is in the same session (Section 7).
7	**Capture effects and missing data.** Capture is not instantaneous (dllhost.exe is stamped one second after the image time) and FTK Imager was running; paged-out registry data meant TimeZoneInformation values and ProfileList could not be read.
8	**Profile choice.** Win7SP1x86_24000 was not run for comparison. The brief's "Windows 10" wording conflicts with the image, which is Windows 7.
9	**Strings are undated and their counts are not events.** Variant addresses (196.128.56.5, 192.128.56.5) occur and are unverified; they are not used.
10	**Scope.** Only the memory image was examined. No disk image, event log, HTTP server log or the USB device itself was available, so the file's content, hash and any copy on F: could not be checked.

Suggested further work (out of scope here). Acquire the disk image and check the NTFS $MFT and USN journal for secret_file.docx in Downloads and on the USB volume; examine Windows event logs and the SYSTEM Enum\USBSTOR timestamps in the on-disk hive; obtain the HTTP server logs for 192.168.56.5 through proper authority; and check the hypervisor logs to confirm the clock step.

6. Timeline (UTC)

All times are UTC on 2019-01-06 and come from the system clock stored in the image, using only values stamped after the early-boot clock step (Section 5.2). Rows describe what the artefact **shows**; the interpretation and its limits are in Appendix C. Where an artefact has no time, the row says so.

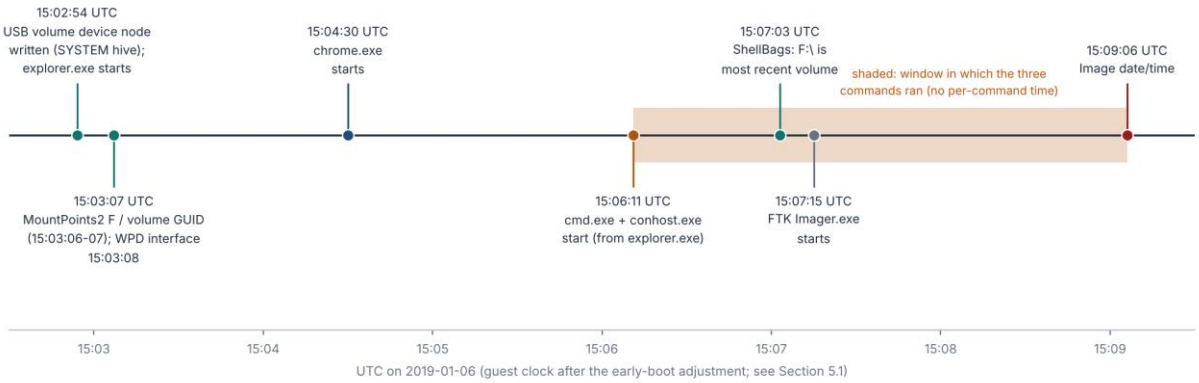


Figure T1*UTC timeline of the observed events (registry key times are last-write times; process times are creation times).*

Stage	UTC timestamp	Evidence ID	Observed fact
Clock context	18:02:40–18:02:45 (early clock)	E03, E05, E15	First processes and the creation time inside volume GUID {421e7e52-…} carry a clock about three hours ahead; the next processes are stamped 15:02:47.
Removable-device evidence	15:02:54	E10, E11	USB device and volume keys (Enum\USBSTOR, Enum\STORAGE\Volume), DeviceClasses and MountedDevices last written; explorer.exe starts.
Removable-device evidence	15:03:06–15:03:08	E12, E10	MountPoints2\F (15:03:06) and volume GUID (15:03:07) created in IEUser's hive; portable-device interfaces for the USB volume created (15:03:08).
Browser or process evidence	15:04:30	E05	chrome.exe 2388 starts in IEUser's session. File-name strings and closed port-80 connections to 192.168.56.5 are undated (E06, E14).
Command-history evidence	15:06:11	E05, E08, E09	cmd.exe 3996 and conhost.exe 3988 created by explorer.exe; UserAssist agrees.
Download-folder review; command-history evidence	between 15:06:11 and 15:09:06 (no per-command time)	E07	ipconfig, cd Downloads, copy secret_file.docx F:; console shows 1 file(s) copied.
Drive mapping	15:07:03	E13	ShellBags: F:\ is the most recently used volume (key last updated).
Capture	15:07:15	E05, E09	FTK Imager.exe 2596 starts (High integrity, IEUser).
Capture	15:09:06	E02	Image date and time; dllhost.exe is stamped 15:09:07.
Supported conclusion	15:06:11–15:09:06	E01–E15	Consistent with secret_file.docx being copied from IEUser's Downloads folder to USB drive F: from an interactive console. Content, intent, legality and the person are not established.

7. Attribution assessment

Question: can the memory evidence attribute the copy to a particular person? **Answer:** it attributes the activity to the **IEUser account session**, and not to a named individual.

Evidence linking the activity to the IEUser session	Evidence and limits against a personal attribution
cmd.exe token user SID …-1000 = IEUser; prompt C:\Users\IEUser\Downloads> (E04, E07)	A SID identifies a profile. Nothing in memory shows who knew the password or sat at the machine.
cmd.exe parent is explorer.exe 2524 in session 1; token has Interactive (S-1-5-4) and Console Logon (S-1-2-1) and no Network group (E08)	An OpenSSH server (sshd, account sshd_server) was listening on port 22 and its profile was loaded (E04, E06). The session evidence argues against a remote shell, but cannot exclude another person using the console or a remote-console tool.
UserAssist in IEUser's hive records cmd.exe launched at 15:06:11 (E09); ShellBags in IEUser's hive record F:\ (E13)	IEUser is a local account present since the 2015 image build and looks like a shared or default lab account.
The USB drive was mounted for IEUser by 15:03:07 (E12)	FTK Imager.exe ran under the same account (E05, E09), so the examiner's own activity is in the same session. The USB device's ownership is unknown.

Conclusion on attribution. The commands were run in an interactive session of the local account IEUser. Naming a person would need evidence outside memory: authentication and event logs, physical-access records, the USB device's history, and interviews. **Confidence in identifying a person: low.**

8. Appendices

Appendix A: screenshots. Appendix B: selected Volatility output. Appendix C: timeline table. Appendix D: evidence log. Appendix E: commands run.

Appendix AScreenshots

All 31 screenshots were taken from the examiner's Kali terminal during the examination. The table lists every screenshot and where it appears. The key evidence screenshots are placed beside the finding in Section 5; the supporting screenshots follow the table.

Figure	What it shows	Evidence ID	Placed in
S01	Case folder structure, first log entry, file located; failed first copy (placeholder path)	E01	Appendix A
S02	Read-only copy, size, SHA-256 of original and working copy, log entry	E01	Appendix A
S03	Volatility 2.6.1 cloned; --info import notices	E02	Appendix A
S04	First imageinfo run started (header)	E02, E16	Appendix A
S05	First imageinfo run killed (zsh: killed)	E02	Appendix A
S06	imageinfo result: profile suggestions, image and local time	E02, E03	Section 5.2
S07	pycryptodome installed for Python 2	E02	Appendix A

Figure	What it shows	Evidence ID	Placed in
S08	pslist start, kdbgscan, plugin list	E02, E03, E05	Section 5.2
S09	Error output, TimeZoneInformation last write, local time, ProfileList re-run	E03, E04	Appendix A
S10	getsids tokens; empty time-zone and ProfileList runs	E04, E08	Section 5.3
S11	SAM account names; cmd.exe token user SID	E04, E08	Section 5.3
S12	pslist later part (15:02:47–15:09:07)	E05	Section 5.4
S13	netscan UDP and listening sockets; hivelist	E04, E06	Appendix A
S14	netscan end (CLOSED rows, sshd port 22); hivelist	E04, E06	Section 5.4
S15	netscan rows, hivelist hive paths, log entry	E04, E06, E16	Appendix A
S16	consoles screen buffer (ipconfig, cd Downloads, copy)	E07	Section 5.5
S17	Console end, netscan state filter, psscan rows	E05, E06, E07	Appendix A
S18	cmdscan history and start of consoles	E07, E08	Section 5.5
S19	Console owner lines, pslist rows, BagMRU volume list	E07, E08, E13	Section 5.5
S20	userassist entries	E09	Section 5.5
S21	DeviceClasses disk class; start of MountedDevices	E10, E11	Section 5.6
S22	MountedDevices values (floppy, CD-ROM, volume GUIDs)	E11	Appendix A
S23	GUID creation-time decode; \DosDevices\F: value	E11, E15	Section 5.6
S24	hivedump SYSTEM: WPD interfaces and Enum\STORAGE\Volume times	E10	Appendix A
S25	hivedump SYSTEM: Enum\USBSTOR device subtree times; ProfileList re-run	E10, E04	Section 5.6
S26	MountPoints2 subkeys (IEUser NTUSER.DAT)	E12	Appendix A
S27	hivedump NTUSER: MountPoints2 key times	E12	Section 5.6
S28	shellbags output end; HTML output file written	E13	Appendix A
S29	ShellBags F:\ and Downloads entries; log entry	E13, E16	Section 5.7
S30	filescan tail and strings results (ASCII and UTF-16 lists)	E14	Appendix A
S31	Remaining strings results; log entry (first attempt failed)	E14, E16	Appendix A

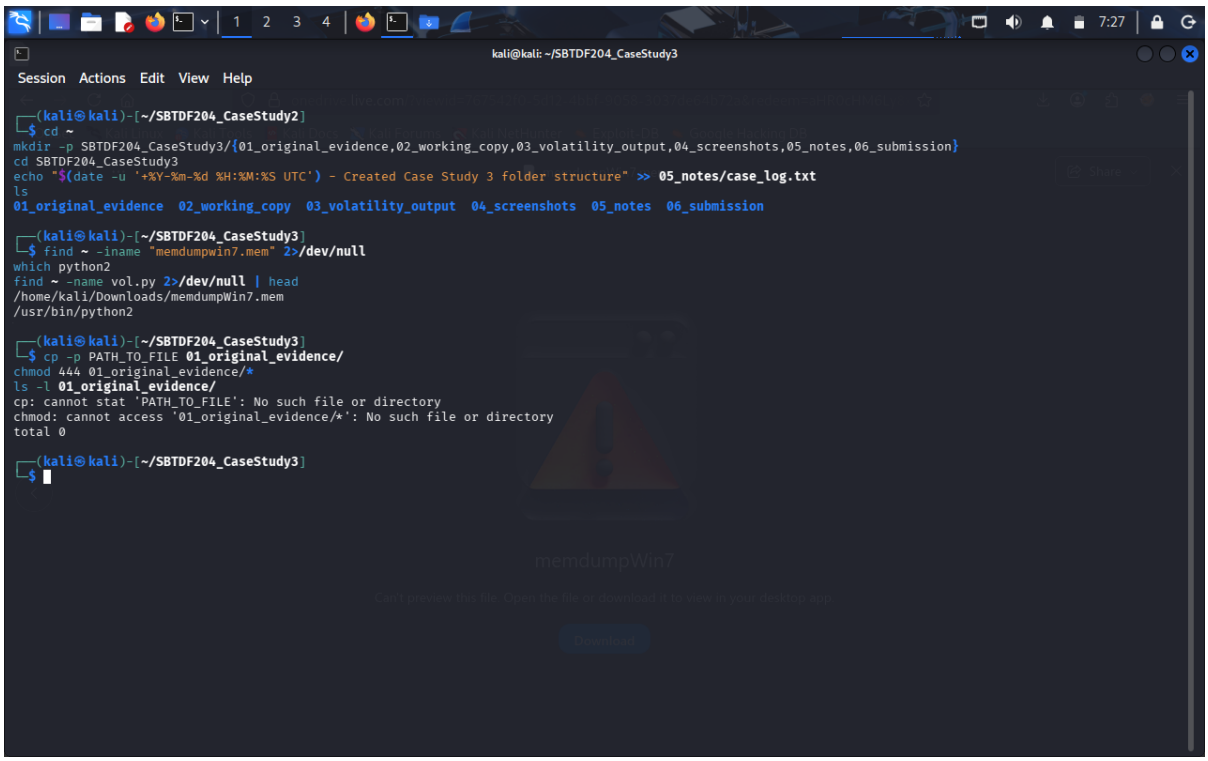


Figure S01*Case folder structure, first case-log entry,* find *locating* memdumpWin7.mem *in* ~/Downloads*; the first copy attempt used a placeholder (*PATH_TO_FILE*) and failed, so nothing was copied (*total 0*). (E01)*

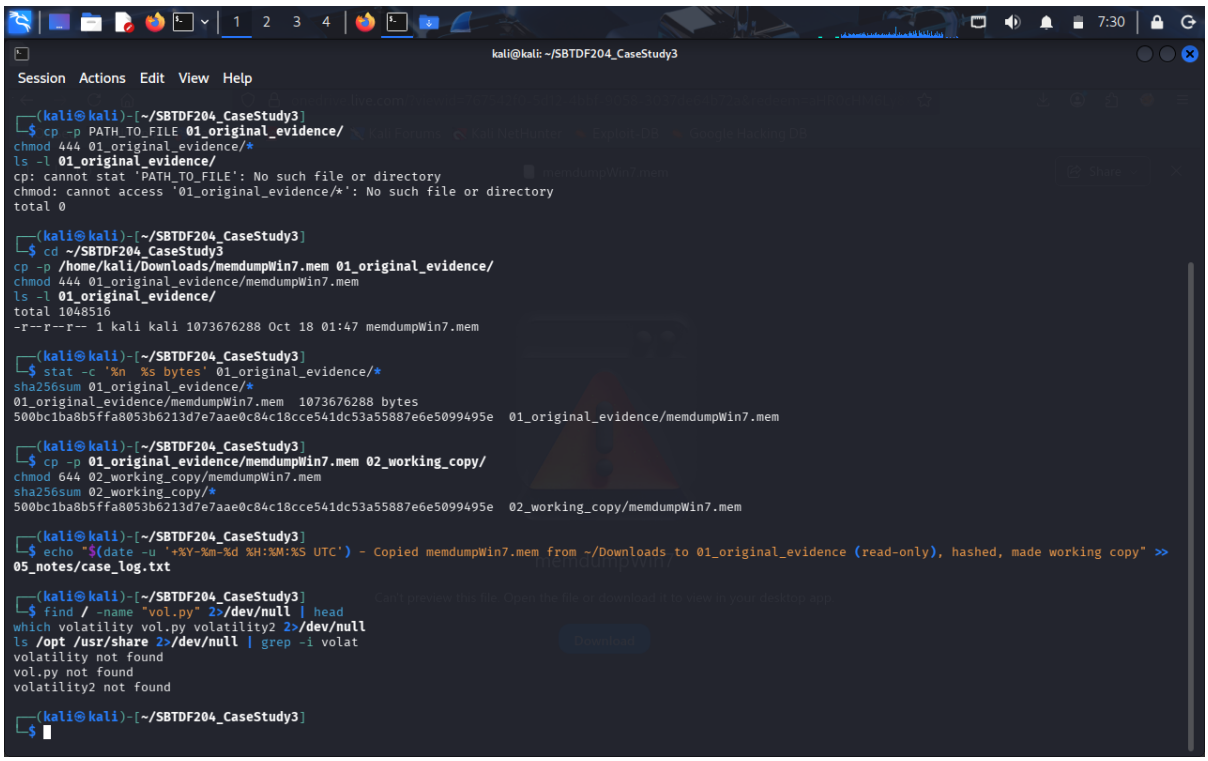


Figure S02*Original copied with* cp -p*, mode 444, 1,073,676,288 bytes; SHA-256 of original and working copy identical (*500bc1ba…5099495e*); log entry;* vol.py *not yet installed. (E01)*

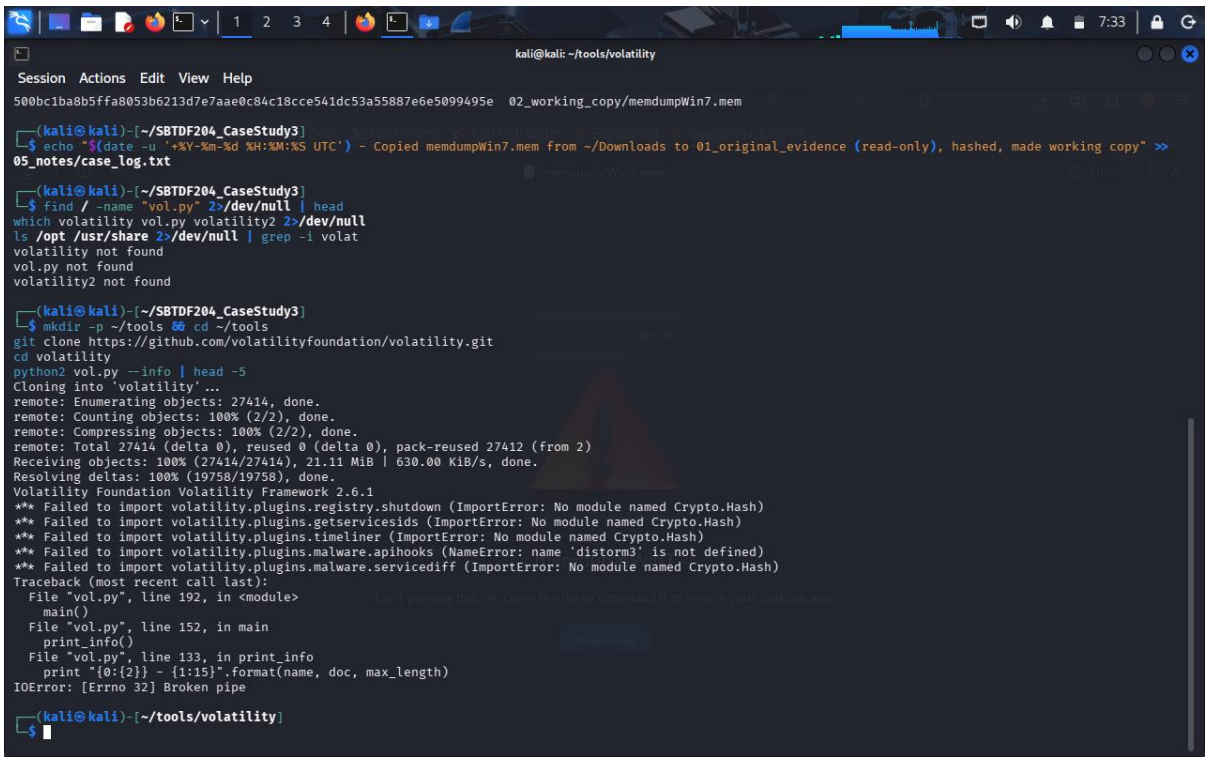


Figure S03*Official Volatility 2.6.1 repository cloned to* ~/tools/volatility*;* python2 vol.py --info *with import notices for* Crypto.Hash *and* distorm3. *(E02)*

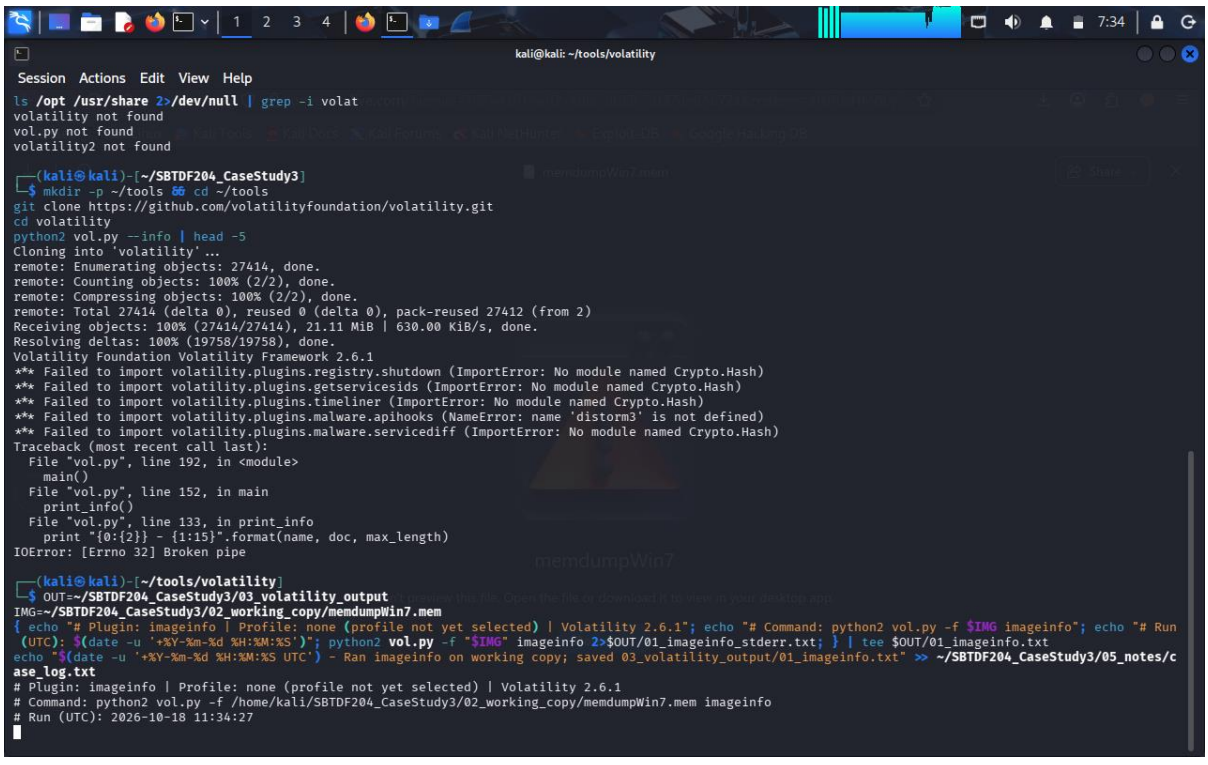


Figure S04*First* imageinfo *run on the working copy started; header (plugin, command, UTC run time) written to* 01_imageinfo.txt. *(E02, E16)*

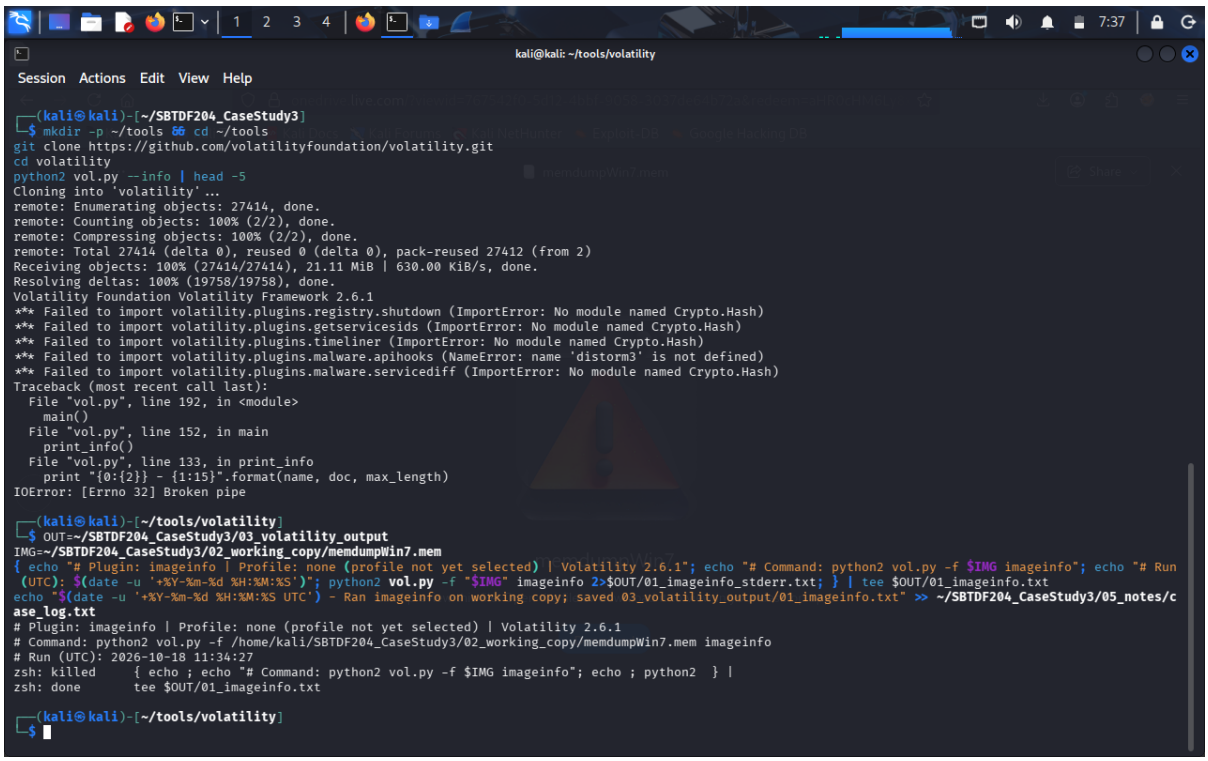


Figure S05*First* imageinfo *run ended with* zsh: killed*;* 01_imageinfo.txt *holds the header only and was not used. (E02)*

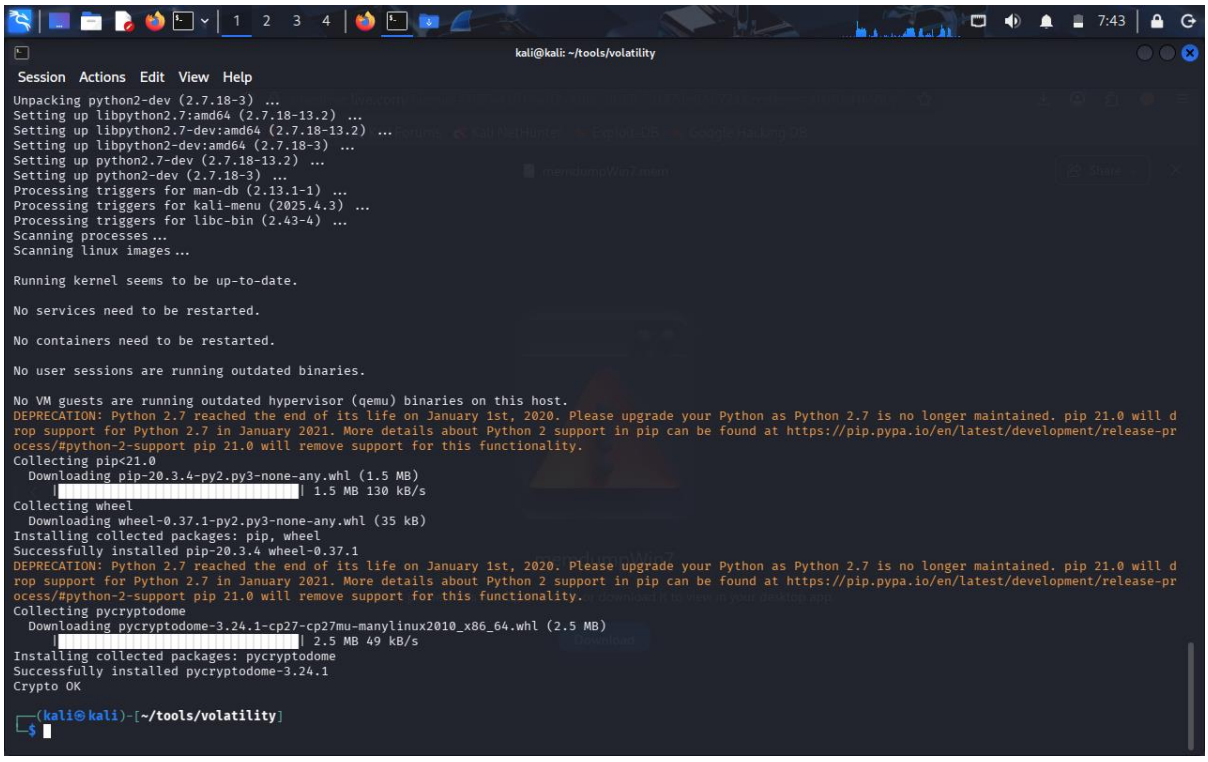


Figure S07pycryptodome *3.24.1 installed for Python 2 (*Crypto OK*), enabling* shellbags, userassist *and* getsids. *(E02)*

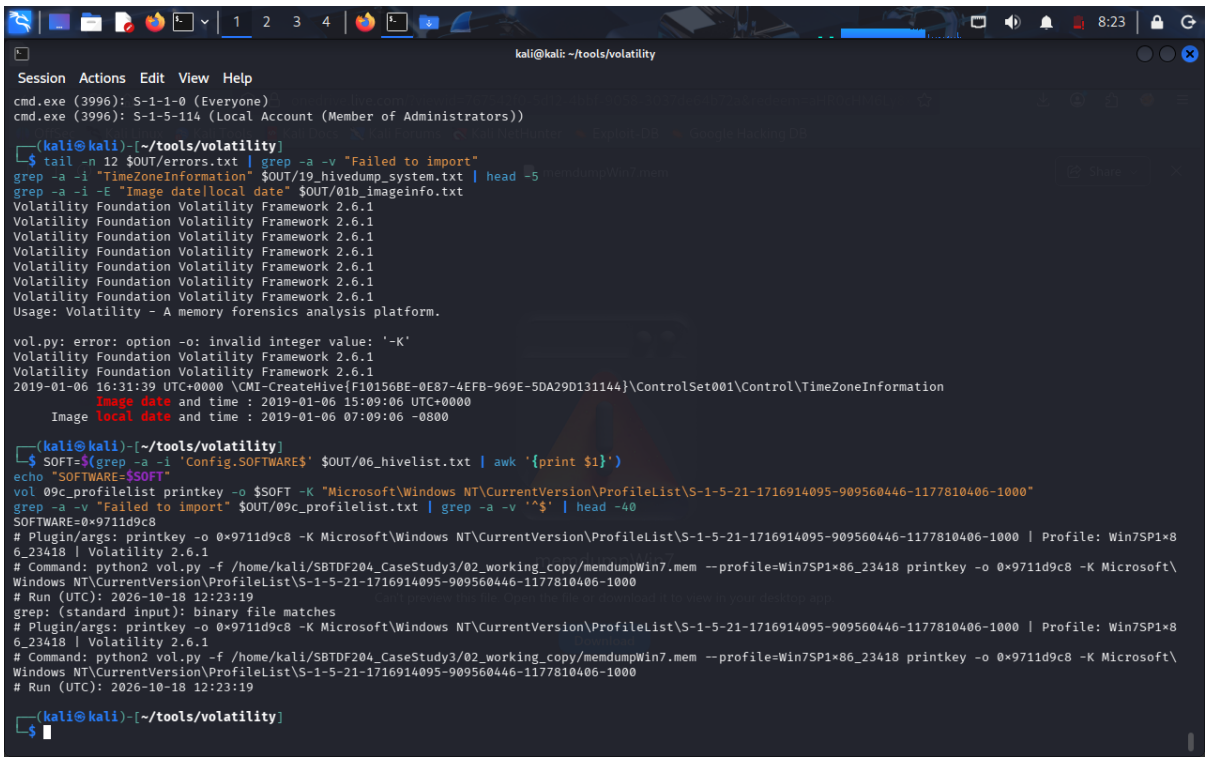


Figure S09errors.txt *(the* -o *option received no value),* hivedump *line for* TimeZoneInformation *(last written 16:31:39 UTC),* imageinfo *image and local date, and the* ProfileList *re-run header. (E03, E04)*

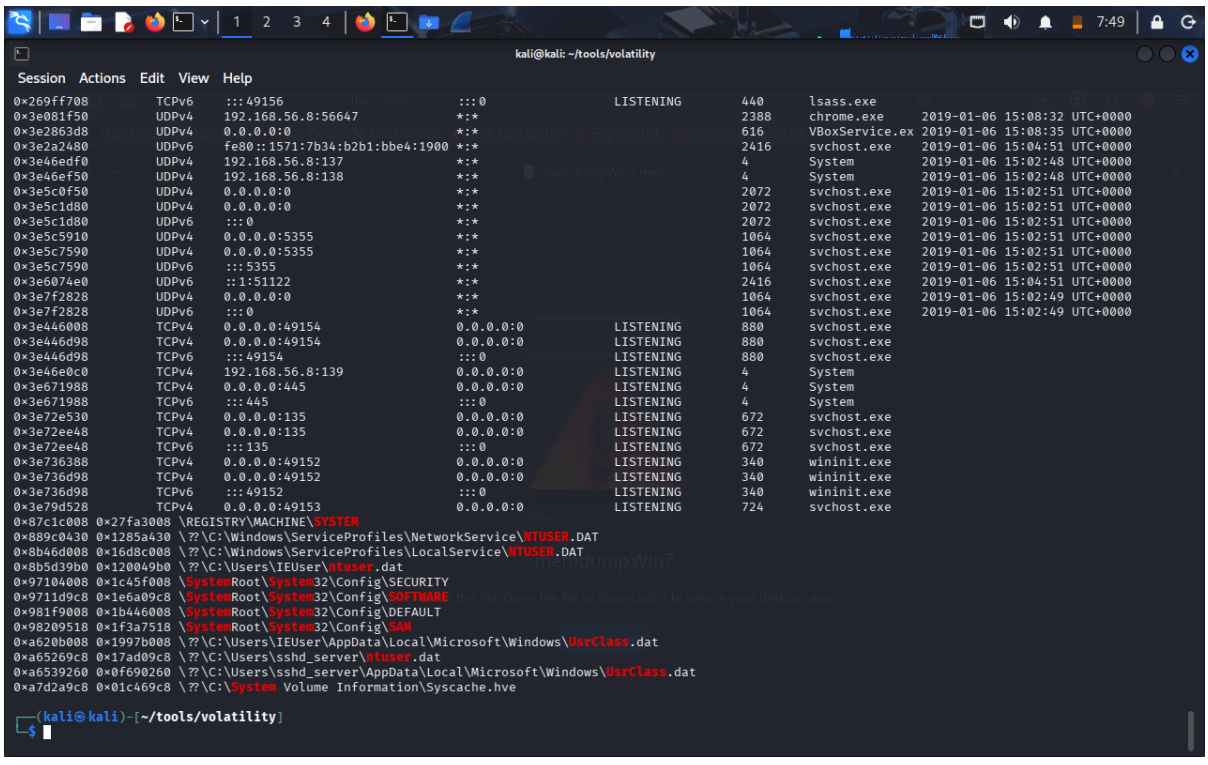


Figure S13netscan *UDP and listening sockets (including* chrome.exe *UDP at 15:08:32 and* VBoxService.exe *UDP at 15:08:35) and* hivelist. *(E04, E06)*

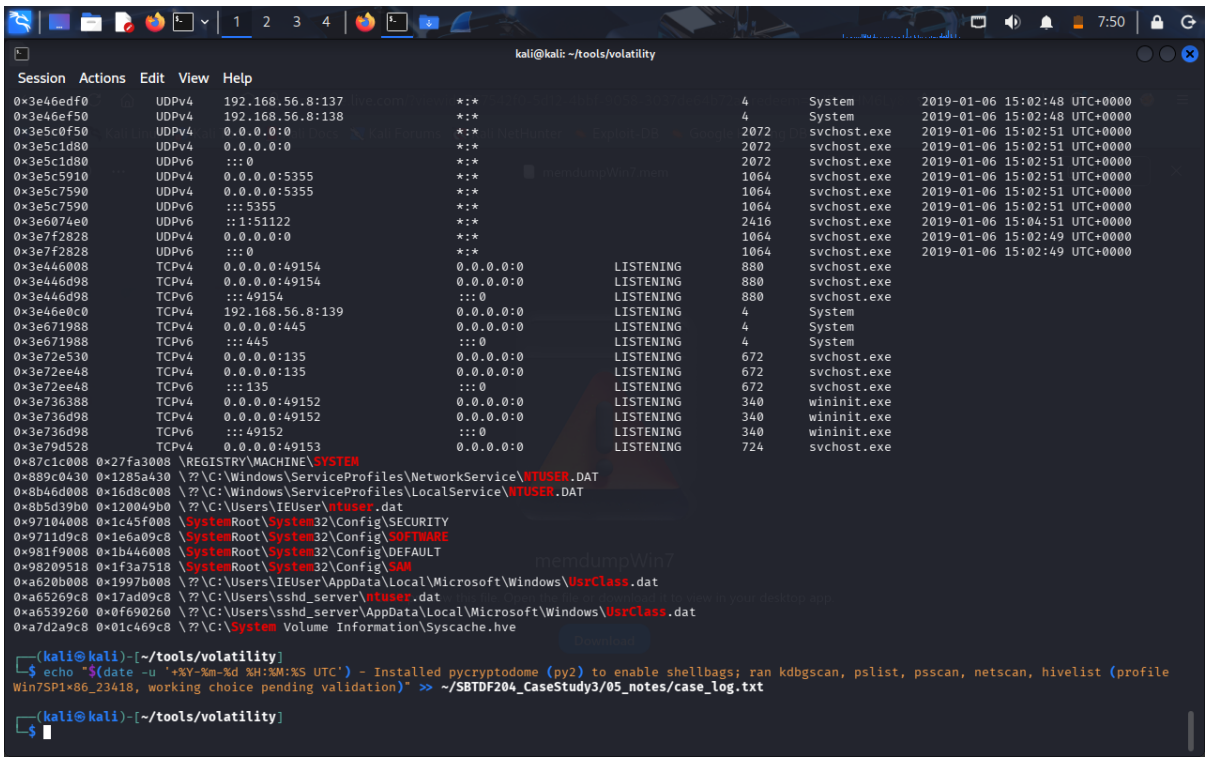


Figure S15netscan *rows,* hivelist *hive paths and the case-log entry for the plugins run to that point. (E04, E06, E16)*

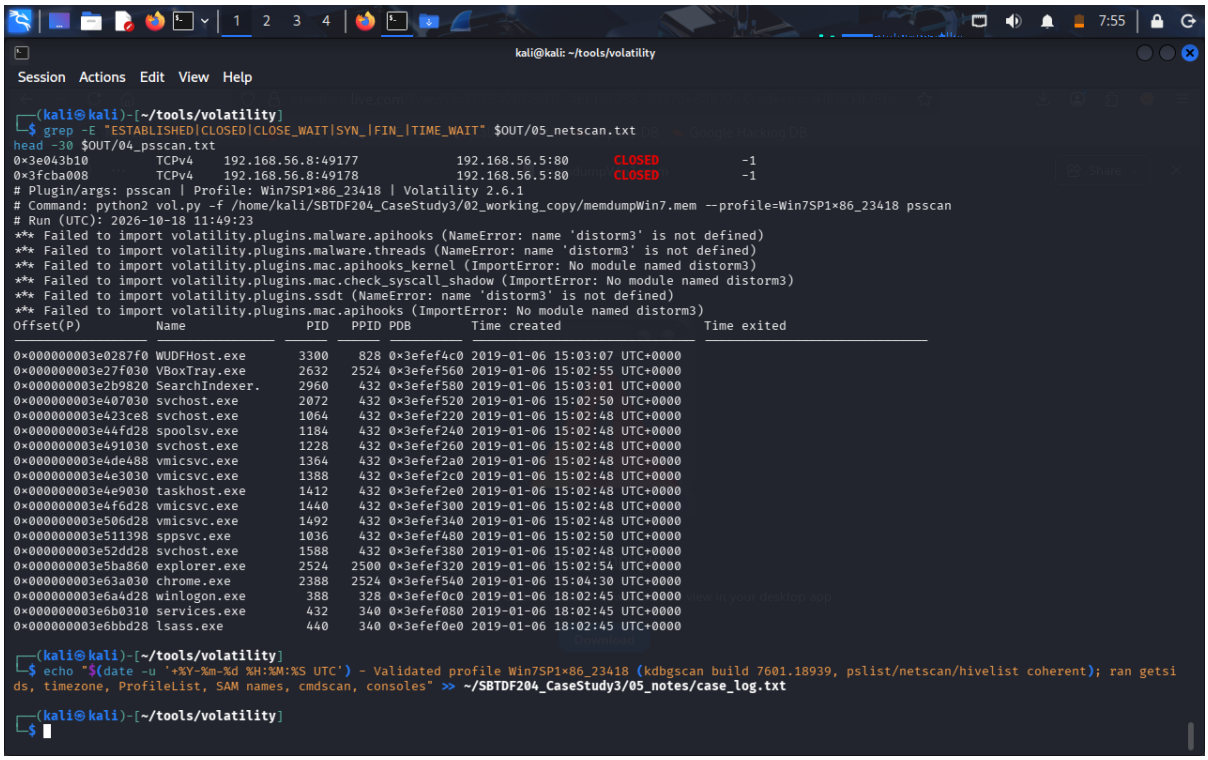


Figure S17*End of the console buffer;* grep *of* netscan *states (only two CLOSED rows, no ESTABLISHED); first* psscan *rows with creation times; log entry follows. (E05, E06, E07)*

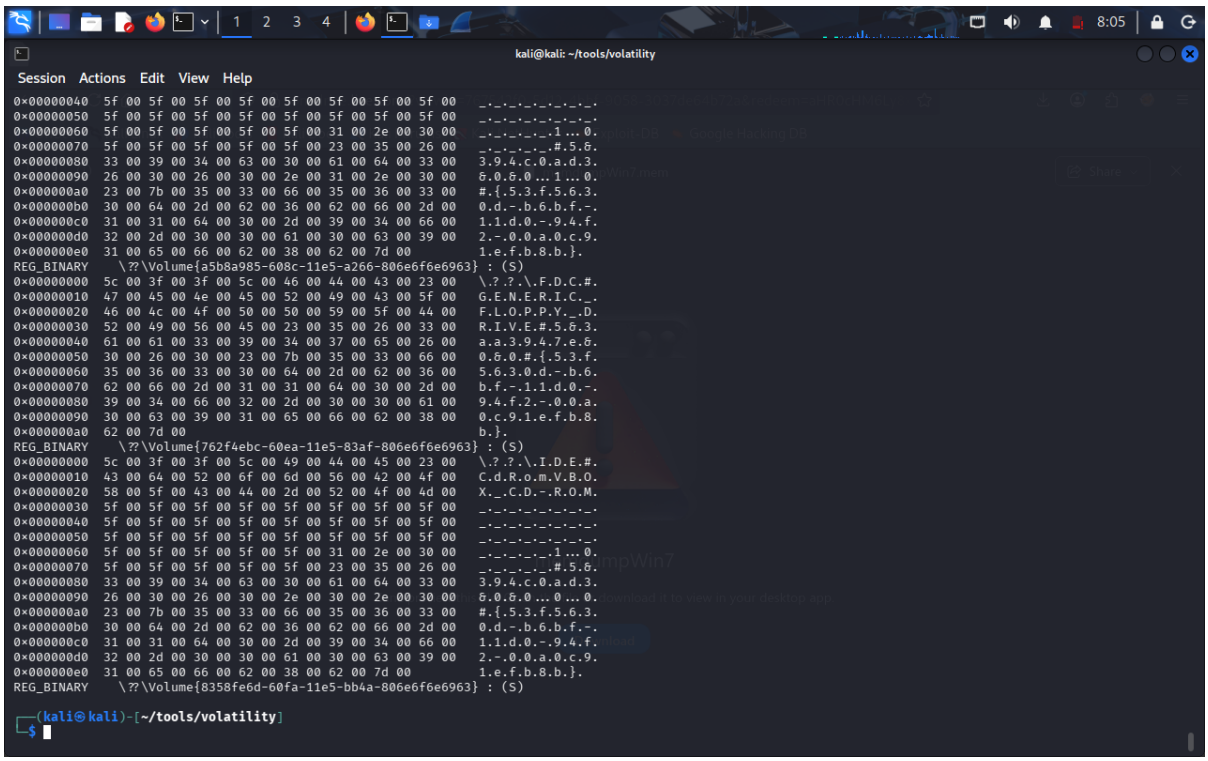


Figure S22MountedDevices *values: floppy (*FDC#GENERIC_FLOPPY_DRIVE*), CD-ROM (*IDE#CdRomVBOX_CD-ROM*) and volume GUID entries (hex and UTF-16 dump). (E11)*

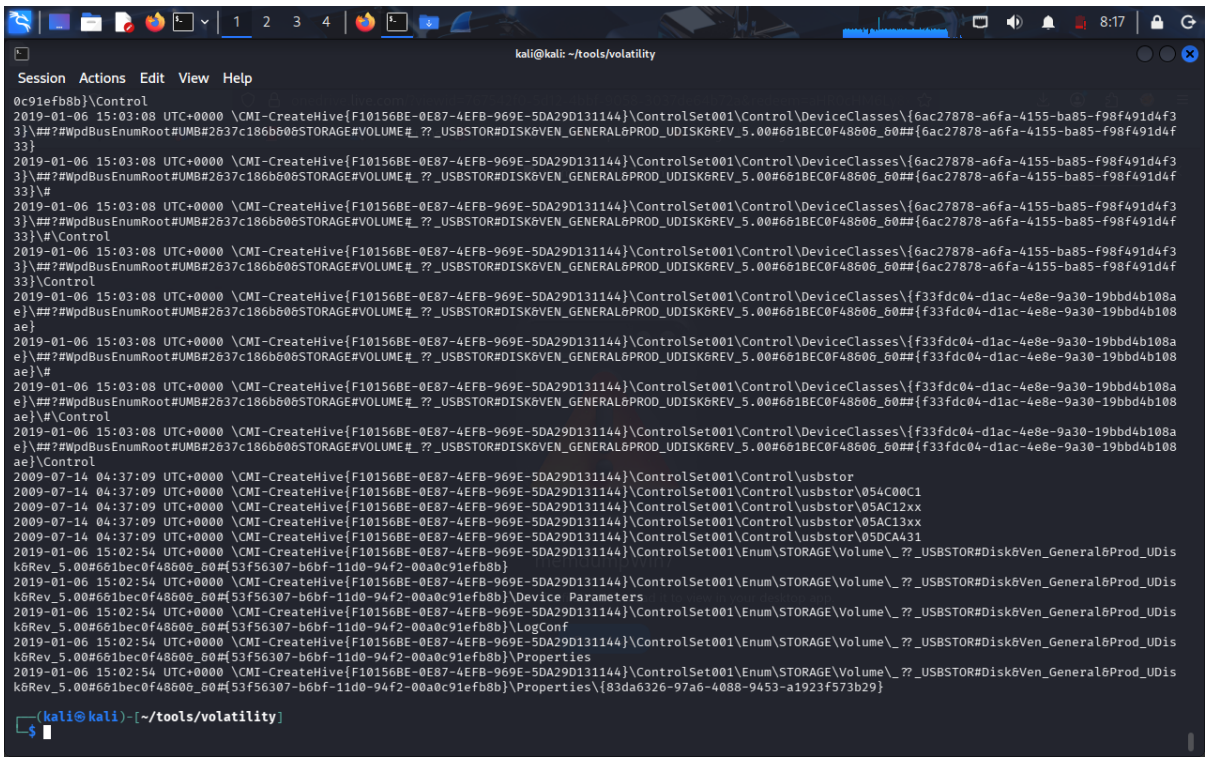


Figure S24hivedump *(SYSTEM) filtered for* usbstor*: WPD interfaces for the USB volume (15:03:08), default* Control\usbstor *entries (2009) and* Enum\STORAGE\Volume_??_USBSTOR#… *keys (15:02:54). (E10)*

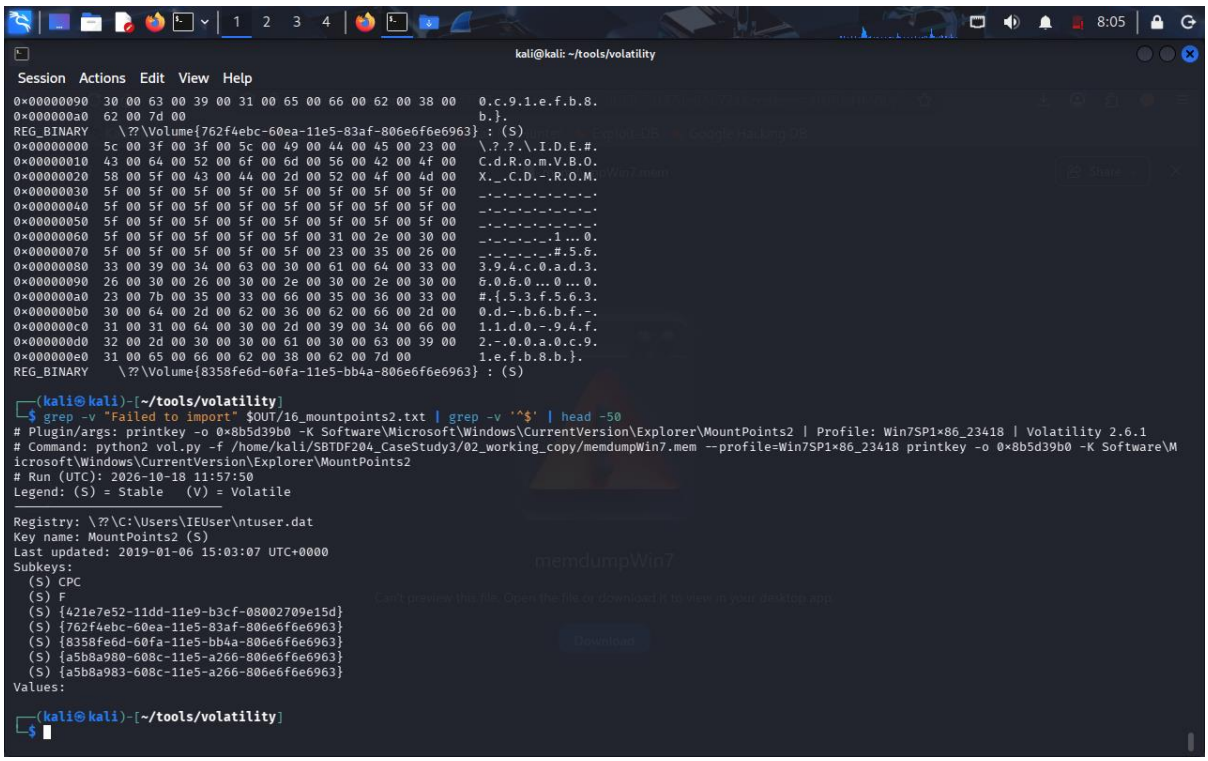


Figure S26MountPoints2 *in IEUser's* NTUSER.DAT *(last updated 15:03:07): subkeys* CPC, F *and five volume GUIDs. (E12)*

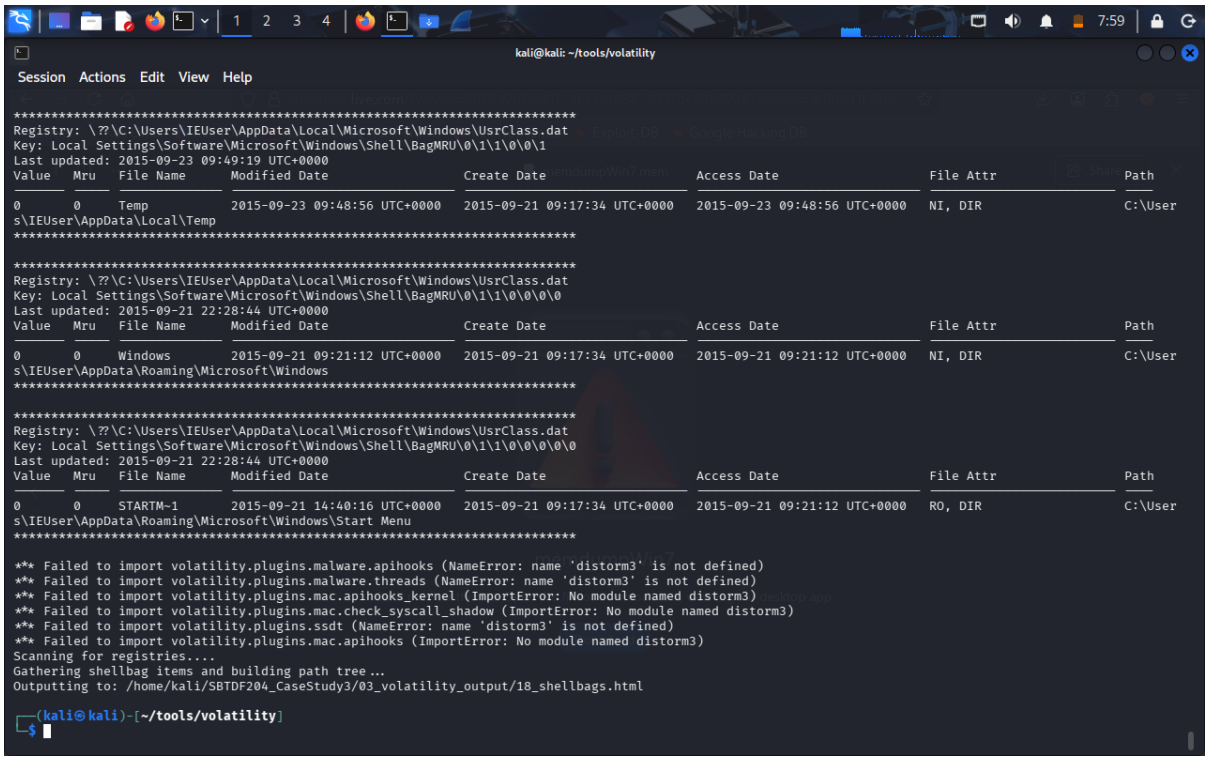


Figure S28*End of the* shellbags *output (BagMRU entries* Temp, Windows, Start Menu *with 2015 dates) and the HTML output file* 18_shellbags.html *written. (E13)*

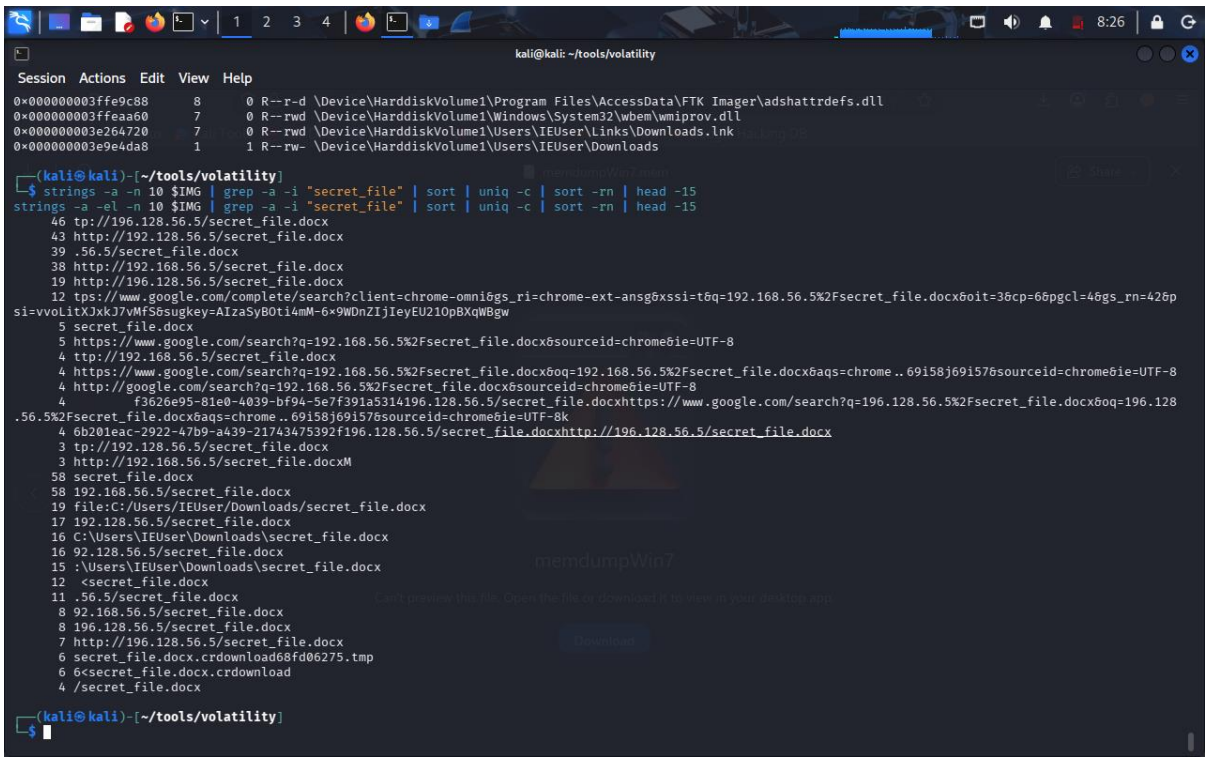


Figure S30filescan *tail (Downloads directory object,* Downloads.lnk*) and the* strings *lists for* secret_file *(ASCII, then UTF-16). (E14)*

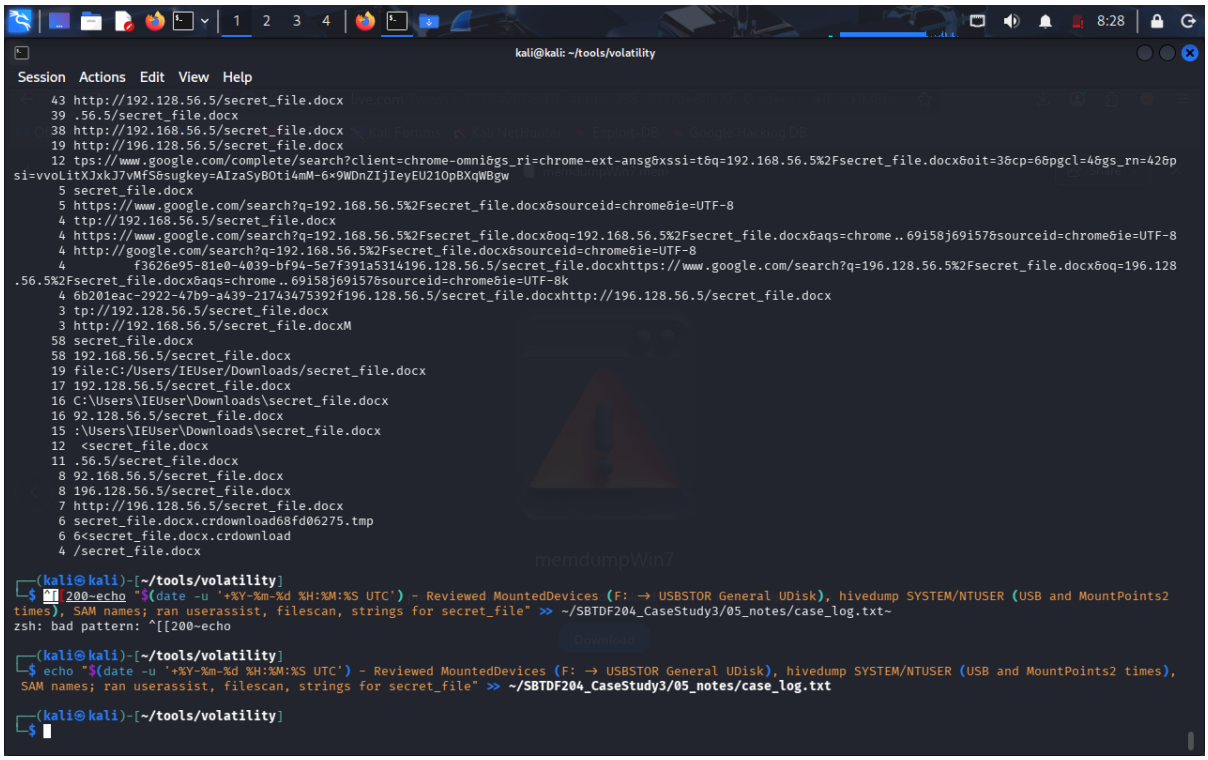


Figure S31*Scrolled view of the same* strings *lists and the case-log entry; the first attempt failed with* zsh: bad pattern *because of pasted control characters and ran nothing, and the second wrote the entry. (E14, E16)*

Appendix BSelected Volatility output

Excerpts are transcribed from the saved output files (03_volatility_output/). Each file begins with # Plugin/args, # Command and # Run (UTC) lines. Long lines were shortened with "…" and a blank line separates excerpts.

B1. imageinfo (profile: none yet; file 01b_imageinfo.txt)

```
Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
        AS Layer1 : IA32PagedMemoryPae (Kernel AS)
        AS Layer2 : FileAddressSpace (/home/kali/SBTDF204_CaseStudy3/02_working_copy/memdumpWin7.mem)
         PAE type : PAE      DTB : 0x185000L      KDBG : 0x82972c68L
  Number of Processors : 1     Image Type (Service Pack) : 1
    KPCR for CPU 0 : 0x82973d00L    KUSER_SHARED_DATA : 0xffdf0000L
  Image date and time : 2019-01-06 15:09:06 UTC+0000
Image local date and time : 2019-01-06 07:09:06 -0800
```

B2. pslist (profile Win7SP1x86_23418), rows used in Section 5.4

```
Offset(V)   Name            PID  PPID Thds Hnds Sess Wow64 Start
0x85bba860  explorer.exe   2524  2500   32  952    1     0 2019-01-06 15:02:54 UTC+0000
0x8583a030  chrome.exe     2388  2524   33  819    1     0 2019-01-06 15:04:30 UTC+0000
0x8431fb08  cmd.exe        3996  2524    1   22    1     0 2019-01-06 15:06:11 UTC+0000
0x842ddd28  conhost.exe    3988   348    2   52    1     0 2019-01-06 15:06:11 UTC+0000
0x84312030  FTK Imager.exe 2596  2524   14  358    1     0 2019-01-06 15:07:15 UTC+0000
0x850d5a38  dllhost.exe    2228   552    6  n/a    1     0 2019-01-06 15:09:07 UTC+0000
```

B3. netscan, connection rows

```
Offset(P)   Proto  Local Address       Foreign Address     State    Pid
0x3e043b10  TCPv4  192.168.56.8:49177   192.168.56.5:80     CLOSED    -1
0x3fcba008  TCPv4  192.168.56.8:49178   192.168.56.5:80     CLOSED    -1
```

B4. consoles owner lines (file 12_consoles.txt)

```
ConsoleProcess: conhost.exe Pid: 3988
OriginalTitle: %SystemRoot%\system32\cmd.exe
AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c


ConsoleProcess: conhost.exe Pid: 2000
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0x54
```

B5. printkey MountedDevices, value \DosDevices\F: (hex dump decoded as UTF-16, truncated at 0x80 in the screenshot)

```
REG_BINARY    \DosDevices\F: : (S)
5f 00 3f 00 3f 00 5f 00 55 00 53 00 42 00 53 00 ...  ->  _??_USBS
... ->  _??_USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f56
```

B6. ShellBags, BagMRU\0 (IEUser UsrClass.dat, last updated 2019-01-06 15:07:03 UTC) and the Downloads item

```
Value  Mru  Entry Type    Path
1      1    Volume Name   C:\
0      4    Volume Name   A:\
3      3    Volume Name   D:\
```

```
2    2   Volume Name   E:\
4    0   Volume Name   F:\


0  0  Folder  374de290-123f-4565-9164-39c4925e467b  Downloads  EXPLORER
1    0    DOWNLO~1    Modified  2019-01-06  14:49:40    Created  2015-09-21  09:17:34    Accessed  2019-01-06  14:49:40    RO,  DIR
C:\Users\IEUser\Downloads
```

B7. userassist (IEUser)

```
%windir%\system32\cmd.exe                          Count: 10  Focus Count: 36  Last updated: 2019-01-06 15:06:11
C:\Users\IEUser\Desktop\AccessData FTK Imager3-2-0.exe      Count: 1   Focus Count: 0   Last updated: 2019-01-06 15:03:29
%ProgramFiles%\AccessData\FTK Imager\FTK Imager.exe        Count: 1   Focus Count: 1   Last updated: 2019-01-06 15:07:15
C:\Users\Public\Desktop\AccessData FTK Imager.lnk          Count: 1   Focus Count: 0   Last updated: 2019-01-06 15:07:15
```

B8. hivedump key times used in Section 5.6

```
2019-01-06 15:02:54  \ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0
2019-01-06                                                
```

Appendix CTimeline table (UTC, 2019-01-06)

Stages follow the brief's template. Times are from the system clock in the image; process times are creation times and registry times are key last-write times. Items without a time are marked.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Clock context	18:02:40–18:02:45 (early clock)	E03, E05, E15	pslist, volume GUID decode (S08, S23)	System through VBoxService.exe (PID 616) stamped 18:02:44–45; GUID {421e7e52-…} decodes to 18:02:40.6; the next processes are stamped 15:02:47.	The early clock ran about 3 h ahead and then stepped back (hypothesis: guest clock synchronisation). Not used to order events.
Removable-device evidence	15:02:54	E10, E11	hivedump SYSTEM, DeviceClasses, MountedDevices (S21, S24, S25)	Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0, Enum\STORAGE\Volume_??_USBSTOR#…, DeviceClasses and MountedDevices keys last written 15:02:54.	A USB mass-storage device was registered by 15:02:54, the same second explorer.exe was created. Last-write time, not plug-in time; the instance ID is not a serial number.
Removable-device evidence	15:03:06–15:03:07	E12	MountPoints2 in IEUser NTUSER.DAT (S26, S27)	MountPoints2\F created 15:03:06; {421e7e52-…} (with shell\Autoplay) and CPC\Volume\{421e7e52-…} created 15:03:07.	IEUser's Explorer saw the new volume about 13 s after starting. The 2019 GUID is linked to F: by timing only (its MountedDevices data was not captured).
Removable-device evidence	15:03:08	E10	hivedump SYSTEM DeviceClasses (S24)	Windows Portable Devices interfaces (WpdBusEnumRoot#UMB…STORAGE#VOLUME#_??_USBSTOR#DISK&VEN_GENERAL&PROD_UDISK…) created.	The USB volume was registered as portable storage by 15:03:08, about 3 min before cmd.exe started.
Drive mapping	key last write 15:02:54; value undated	E11	MountedDevices \DosDevices\F: (S21–S23)	Value reads _??_USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f56….	Drive letter F: was assigned to the USB device. The value has no time of its own.
Capture tool (context)	15:03:29; 15:03:34	E09, E05	UserAssist; pslist (S20, S12)	AccessData FTK Imager3-2-0.exe last updated 15:03:29 (count 1); msiexec.exe 3748 created 15:03:34.	FTK Imager installer launched; msiexec.exe is consistent with installation (inference). Not part of the transfer.
Browser or process evidence	15:04:30	E05	pslist, psscan (S12, S17)	chrome.exe 2388 created (parent explorer.exe 2524, session 1); children 1280, 2380 (15:04:30), 2912 (15:04:32), 976 (15:06:23).	Chrome was running in IEUser's session before cmd.exe started. Process start time, not a page-visit time.
Browser or process evidence	undated	E14	strings on the working copy (S30, S31)	http://192.168.56.5/secret_file.docx; Chrome omnibox search URLs for 192.168.56.5/secret_file.docx; secret_file.docx.crdownload; C:\Users\IEUser\Downloads\secret_file.docx.	Consistent with Chrome requesting the file from 192.168.56.5 and saving it in Downloads. Strings have no time; counts are memory remnants; variant addresses (196.128…, 192.128…) are unverified.
Browser or process evidence	undated	E06	netscan (S14, S17)	Two CLOSED TCP connections 192.168.56.8:49177 and :49178 → 192.168.56.5:80, owner PID −1; no ESTABLISHED connections.	Same host and HTTP port as the string URL, but owner unidentified and no time; not tied to Chrome. 192.168.56.5 was not contacted.
Command-history evidence	15:06:11	E05, E08, E09	pslist, consoles, UserAssist (S12, S19, S20)	cmd.exe 3996 and conhost.exe 3988 created; parent explorer.exe 2524, session 1; token has Interactive and Console Logon SIDs; UserAssist cmd.exe last updated 15:06:11.	The console was opened from IEUser's Explorer. This starts the window in which the commands ran.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Download-folder review; command-history evidence	between 15:06:11 and 15:09:06 (no per-command time)	E07	cmdscan, consoles (S16–S18)	History: ipconfig (#0), cd Downloads (#1), copy secret_file.docx F: (#2, LastAdded: 2). Screen: C:\Users\IEUser\Downloads>copy secret_file.docx F: then 1 file(s) copied.	Order is known and the copy was the last command; success is screen text only. No file object on F: was recovered. History and screen buffer are one console seen twice.
Download-folder review	snapshot 14:49:40 (earlier session)	E13	ShellBags item DOWNLO~1 (S29)	Downloads folder Modified and Accessed 14:49:40, Created 2015-09-21 09:17:34, path C:\Users\IEUser\Downloads.	A folder snapshot taken at an earlier Explorer visit; not the time secret_file.docx arrived or was viewed, and earlier than this boot's processes.
Drive mapping (view)	15:07:03	E13	ShellBags BagMRU\0 in IEUser UsrClass.dat (S19, S29)	Key last updated 15:07:03; volumes C:\, A:\, D:\, E:\, F:\; F:\ has MRU 0 (most recent).	IEUser's Explorer navigated to F:\ about 52 s after cmd.exe started. No file names; key time, not a file time.
Capture tool (context)	15:07:15	E05, E09	pslist, UserAssist (S12, S20)	FTK Imager.exe 2596 created (parent 2524, High integrity, IEUser); UserAssist FTK Imager.exe and its shortcut 15:07:15.	The capture tool ran in the same session about 1 min 51 s before the image time; its presence affects memory and attribution.
Capture	15:09:06	E02, E03	imageinfo, pslist (S06, S12)	Image date and time 15:09:06 UTC (07:09:06 −0800); dllhost.exe stamped 15:09:07.	Upper bound for all events. Capture is not instantaneous (a process stamped 1 s later).
Supported conclusion	15:06:11–15:09:06	E01–E15	All of the above	An interactive IEUser cmd.exe session ran copy secret_file.docx F:; the console reported 1 file copied; F: is a USB mass-storage device mounted since about 15:03.	**Moderate confidence.** Consistent with a copy to removable media. Content, intent, legality and the person are not established (Sections 5.8, 7).

Appendix DEvidence log

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E01	cp, chmod, stat, sha256sum	~/Downloads/memdumpWin7.mem → 01_original_evidence/ → 02_working_copy/	1,073,676,288 bytes; SHA-256 500bc1ba8b5f…5099495e for the original and the working copy (identical); original mode 444.	S01, S02; Section 3
E02	Volatility imageinfo, kdbgscan; pip (pycryptodome)	01b_imageinfo.txt, 02_kdbgscan.txt	Profiles Win7SP1x86_23418 etc.; build 7601.18939.x86fre.win7sp1_gdr.15; DTB 0x185000; KDBG 0x82972c68; image time 15:09:06 UTC; first run killed.	S03–S08; Appendix B1
E03	imageinfo; hivedump SYSTEM; pslist	01b_imageinfo.txt, 19_hivedump_system.txt, 03_pslist.txt	Local time 07:09:06 −0800; TimeZoneInformation last written 16:31:39 (values unreadable); early-boot processes 18:02:44–45 vs 15:02:47+.	S06, S08, S09, S12; Section 5.2
E04	getsids, printkey (SAM Names), hivelist	07_getsids.txt, 10_sam_names.txt, 06_hivelist.txt	IEUser SID …-1000; accounts Administrator, Guest, IEUser, sshd, sshd_server; hive paths and offsets; ProfileList empty.	S10, S11, S13–S15, S25; Section 5.3
E05	pslist, psscan	03_pslist.txt, 04_psscan.txt	Process tree and creation times: explorer.exe 15:02:54; chrome.exe 15:04:30; cmd.exe/conhost.exe 15:06:11; FTK Imager.exe 15:07:15; dllhost.exe 15:09:07.	S08, S12, S17; Appendix B2
E06	netscan	05_netscan.txt	Two CLOSED TCP rows to 192.168.56.5:80 (PID −1); no ESTABLISHED; sshd listening on 22; Chrome UDP 15:08:32.	S13–S15, S17; Appendix B3
E07	cmdscan, consoles	11_cmdscan.txt, 12_consoles.txt	History 0x110448 (3 commands, LastAdded: 2) on conhost 3988 / cmd.exe 3996; screen shows 1 file(s) copied.; OpenSSH console with no commands.	S16–S19; Appendix B4
E08	pslist (PPID, session); consoles (AttachedProcess); getsids	03_pslist.txt, 12_consoles.txt, 07_getsids.txt	cmd.exe 3996: parent explorer.exe 2524, session 1; token Interactive + Console Logon, no Network, Medium integrity; FTK Imager.exe High integrity.	S10, S11, S19; Section 5.3
E09	userassist	21_userassist.txt	cmd.exe 15:06:11; FTK Imager installer 15:03:29; FTK Imager.exe and shortcut 15:07:15.	S20; Appendix B7
E10	printkey DeviceClasses; hivedump SYSTEM	15_deviceclasses.txt, 19_hivedump_system.txt	USBSTOR General UDisk Rev 5.00 6&1bec0f48&0&_&0; Enum and Volume keys 15:02:54; WPD interfaces 15:03:08.	S21, S24, S25; Appendix B8
E11	printkey MountedDevices	14_mounteddevices.txt	\DosDevices\F: → _??_USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#…; key last updated 15:02:54.	S21–S23; Appendix B5
E12	printkey / hivedump MountPoints2	16_mountpoints2.txt, 20_hivedump_ntuser.txt	F created 15:03:06; {421e7e52-…} 15:03:07; key updated 15:03:07; older GUIDs 2015 and 14:32:25.	S26, S27; Appendix B8
E13	shellbags (text and HTML)	17_shellbags.txt, 18_shellbags.html	BagMRU\0 updated 15:07:03; volume F:\ MRU 0; Downloads item with 14:49:40 folder times.	S19, S28, S29; Appendix B6
E14	strings (ASCII and UTF-16); filescan	22_filescan.txt; working copy (strings not saved)	http://192.168.56.5/secret_file.docx; omnibox searches; .crdownload; C:\Users\IEUser\Downloads\secret_file.docx; Downloads directory object.	S30, S31; Section 5.5

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E15	Python 3 uuid (version-1 time)	Volume GUIDs from MountedDevices / MountPoints2	{421e7e52-11dd-11e9-b3cf-08002709e15d} → 2019-01-06 18:02:40.6 UTC; the other four GUIDs → September 2015.	S23; Section 5.6
E16	echo to case_log.txt; output headers	05_notes/case_log.txt	UTC-stamped log of each step; each output file begins with plugin, profile, command and UTC run time.	S02, S15, S29, S31

Appendix ECommands run

$IMG is ~/SBTDF204_CaseStudy3/02_working_copy/memdumpWin7.mem; $PROF is Win7SP1x86_23418; $OUT is ~/SBTDF204_CaseStudy3/03_volatility_output. Each plugin ran as python2 vol.py -f $IMG --profile=$PROF <plugin> with output saved to the named file.

#	Command	Output file	Purpose and result
1	sha256sum, stat, cp -p, chmod 444 on memdumpWin7.mem	terminal; case log	Preserve and hash (Q1)
2	git clone https://github.com/volatilityfoundation/volatility.git; python2 vol.py --info	terminal	Install Volatility 2.6.1
3	python2 vol.py -f $IMG imageinfo	01_imageinfo.txt (killed), 01b_imageinfo.txt	Profile suggestions, image time
4	kdbgscan	02_kdbgscan.txt	Confirm kernel build and profile
5	pslist; psscan	03_pslist.txt; 04_psscan.txt	Process tree and times (Q4)
6	netscan	05_netscan.txt	Sockets and connections (Q4)
7	hivelist	06_hivelist.txt	Hive paths and offsets (Q3, Q6)
8	getsids -p 2524,3996,2388,2596	07_getsids.txt	Process tokens (Q3)
9	printkey -K "ControlSet001\Control\TimeZoneInformation" (also with -o 0x87c1c008)	08_timezone.txt, 08b_timezone.txt	No values returned (Q2)
10	printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileList" (re-run with -o 0x9711d9c8 and the IEUser SID key)	09_profilelist.txt, 09c_profilelist.txt	No data recovered (Q3); the first re-run failed because $SOFT was empty
11	printkey -K "SAM\Domains\Account\Users\Names"	10_sam_names.txt	Local account names (Q3)
12	cmdscan; consoles	11_cmdscan.txt; 12_consoles.txt	Command history and console screen (Q5)
13	printkey -K "ControlSet001\Enum\USBSTOR"	13_usbstor.txt	Output contained non-text bytes; reviewed through hivedump instead (Q6)
14	printkey -K MountedDevices	14_mounteddevices.txt	Drive-letter mappings (Q6)
15	printkey -K "ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}"	15_deviceclasses.txt	Disk devices incl. USBSTOR (Q6)
16	printkey -o 0x8b5d39b0 -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2"	16_mountpoints2.txt	Volumes seen by IEUser (Q6)
17	shellbags; shellbags --output=html --output-file=$OUT/18_shellbags.html	17_shellbags.txt; 18_shellbags.html	Explorer folder and volume history (Q7)
18	hivedump -o 0x87c1c008 (SYSTEM); hivedump -o 0x8b5d39b0 (IEUser NTUSER)	19_hivedump_system.txt; 20_hivedump_ntuser.txt	Key last-write times (Q2, Q6)
19	userassist	21_userassist.txt	Program launch times (Q5)
20	filescan; strings -a -n 10 $IMG and strings -a -el -n 10 $IMG filtered for secret_file; python3 (uuid time of volume GUIDs)	22_filescan.txt; strings and decode not saved	File traces (Q5); GUID creation times (Q2, Q6)